

Golf Launch Monitor Technology Review

**Measurement Principles, Patent Landscape, and Club/Ball
Parameter Estimation**

A comprehensive technical reference

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Abstract

This review surveys the measurement technology behind commercial golf launch monitors — Doppler-radar systems (TrackMan, FlightScope, Garmin R10, Full Swing KIT), photometric camera systems (Foresight GCQuad/GC3, Uneekor, SkyTrak, ProTee VX, Garmin R50), and the hybrid radar+camera architectures that now dominate the high end — with emphasis on *how each system measures or infers club-delivery parameters* (club speed, club path, attack angle, face angle, dynamic loft, impact location) *and ball-launch parameters* (ball speed, launch angles, spin rate, spin axis). Primary sources include the governing patent families (TrackMan/Tuxen radar spin and spin-axis patents, the FlightScope/EDH dielectric-lens spin patent, the Wintriss/Foresight photometric patents, and the Acushnet stereo launch-monitor lineage), regulatory (FCC) filings, manufacturer technical documentation, and the peer-reviewed validation literature. The review develops the underlying physics — the D-plane impact model, oblique-impact spin generation, gear effect, and golf-ball aerodynamics — as the common mathematical framework for interpreting launch-monitor outputs, and closes with a technical map of patent disclosures and design guidance for implementers building or evaluating a system. The central organizing principle is the provenance of each reported quantity: the observations, calibration and model assumptions that support it. Combining radar and optical observations can constrain different parts of the inference problem, but does not automatically make every output directly measured or independently validated. Early optical disclosures illustrate surface-image rotation and club-marker geometry; their age does not establish implementation accuracy or design-specific rights clearance.

This document is written to be vendor- and project-neutral. Where a specific product is named it is as evidence — a published definition, a measured tolerance, a patent claim — not as an endorsement or a design target.

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Chapter 1

Introduction

1.1 Purpose and Scope

A golf launch monitor answers two questions for every shot: *what did the club do through impact* (delivery), and *what did the ball do after impact* (launch and flight). Commercial systems answer these questions with strikingly different sensor architectures — continuous-wave Doppler radar, multi-camera photogrammetry, and, increasingly, fusions of the two — yet they all implement the same underlying physics: an oblique-impact collision model connecting club delivery to ball launch, and an aerodynamic model connecting ball launch to flight.

This review is a technical reference on how launch monitors work. It is written for engineers building or evaluating such a system, for researchers using one as an instrument, and for anyone who needs to know what a reported number actually represents. Its goals are:

1. catalogue *how each major commercial system works*, at the level of radar physics, imaging geometry, and signal processing;
2. establish, parameter by parameter, *what is measured directly, what is inferred through a model, and what is essentially estimated* — especially for club-delivery data such as face angle and club path;
3. survey the *patent landscape* as a source of disclosed sensor architectures, signal-processing methods and claim limitations, while distinguishing those disclosures from verified commercial implementations and design-specific rights determinations;
4. collect the *physics and algorithms* (D-plane, impact mechanics, gear effect, aerodynamic coefficients, spectral spin estimation, stereo triangulation, Kalman filtering) that any implementation needs; and
5. translate all of the above into *concrete design guidance*, organised by sensing architecture rather than by product.

Key point

A note on neutrality. Products are named throughout, and often criticised. That is because published vendor definitions, patent claims and measured tolerances are the primary evidence available in this field — the peer-reviewed literature is thin, as Chapter 9 documents. Naming a system is a citation, not a recommendation, and no architecture in this review is presented as the one to build.

1.2 The Two Sensing Families

Doppler radar systems (TrackMan, FlightScope, Garmin Approach R10, Full Swing KIT) illuminate the hitting area and downrange volume with a microwave carrier and extract target radial

velocity from the Doppler shift. Multiple receive antennas turn a velocity sensor into a 3D tracker via phase interferometry (Chapter 4). Radar excels outdoors: it tracks the entire flight, so carry and curvature are *observed*, not modeled. Its weaknesses are at the club face — radar cannot see face orientation directly — and indoors, where a screen truncates the observable flight.

Photometric (camera) systems (Foresight GC-series, Uneekor, SkyTrak, ProTee VX, Garmin R50) capture a burst of high-speed, IR-strobed stereo images over the first ~ 30 cm of ball flight and reconstruct position and orientation photogrammetrically (Chapter 5). They measure launch conditions — including 3D spin from dimple-pattern rotation — essentially perfectly for simulator purposes, then *model* the flight. Their weakness is the mirror image of radar’s: everything downrange of the capture volume is simulated, and club data requires either fiducial stickers on the face or an overhead viewing geometry.

The market’s convergent evolution is the single most instructive fact for a new design: *every* high-end vendor has concluded that neither modality suffices alone. TrackMan added cameras to its radar (OERT, Section 4.6); FlightScope added Fusion Tracking cameras; SkyTrak added radar to its camera; Rapsodo pairs radar with impact cameras; Full Swing feeds a camera into its radar ML pipeline. Chapter 6 tabulates the landscape.

1.3 Reading Guide

Chapter 2 fixes terminology and coordinate conventions (TrackMan’s definitions, the de facto industry standard). Chapter 3 develops the impact physics connecting club delivery to ball launch — the D-plane model, spin generation, and gear effect — which every monitor uses either forward (simulation) or inverse (parameter estimation). Chapters 4 and 5 treat the two sensing families in depth. Chapter 6 surveys the commercial devices. Chapter 7 maps the patent landscape with expiry status. Chapter 8 covers ball-flight aerodynamics and trajectory estimation. Chapter 9 reviews the independent validation literature. Chapter 10 distills the review into design guidance.

Key point

Throughout, we use a right-handed coordinate system for a right-handed golfer: x down the target line, y up, z to the golfer’s right. Angles are positive right/up. All club-delivery values are referenced to the moment of *maximum ball compression*; all ball-launch values to the moment of *separation from the face*.

Chapter 2

Parameter Definitions and Conventions

Launch monitors from different vendors disagree partly because they measure different things and partly because they *define* things differently. TrackMan’s definitions [67, 62] are the industry reference and are adopted here. The critical subtlety is *when* each quantity is defined: club-delivery parameters at the time of **maximum compression**, ball parameters **immediately after separation**.

2.1 Club-Delivery Parameters

Table 2.1: Club-delivery parameters (TrackMan conventions). “GC” = the geometric center of the club head.

Parameter	Definition	Notes
Club speed	Linear speed of the GC just prior to first contact	Not the impact-point speed; the toe moves up to ~ 7 mph faster than the heel
Attack angle	Vertical direction of GC motion at maximum compression	+ = hitting up. PGA Tour driver avg $\approx -1.3^\circ$; LPGA $\approx +3^\circ$
Club path	Horizontal direction of GC motion at maximum compression	+ = in-to-out (right of target for RH)
Face angle	Horizontal direction the face normal points, at the center-point of ball contact, at maximum compression	+ = open
Face to path	Face angle – club path	Sign controls curvature
Dynamic loft	Vertical angle of the face normal at the contact point at maximum compression	Differs from static loft via shaft lean/bend, face roll, impact height
Spin loft	3D angle between the club-motion direction (path, attack angle) and the face-normal direction (face angle, dynamic loft)	$\approx$ dynamic loft – attack angle only when face-to-path ≈ 0
Swing plane	Vertical angle of the plane traced by GC motion vs. the horizon	
Swing direction	Horizontal angle of that plane’s base vs. the target line	
Low point	Distance from GC at maximum compression to the lowest point of the swing arc	+ = low point ahead of the ball (ball-first contact)
Impact height / offset	Vertical / horizontal strike location relative to face center	Drives gear effect (Section 3.4)
Dynamic lie	Shaft angle vs. horizontal at impact	
Closure rate	Angular rate at which the face is closing in 3D ($^\circ/\text{s}$)	Foresight GCQuad-class only

Two definitional choices deserve emphasis because they are common sources of inter-device disagreement:

1. **Which point on the club is tracked.** TrackMan defines club speed at the *geometric center* of the head and reconstructs that point from the radar “3D silhouette” of the head [69]. A naive Doppler processor instead locks to the strongest or fastest return — often the toe of a driver — and reports a speed several mph high. This single choice explains much of the chronic club-speed disagreement between brands.
2. **Where on the face orientation is evaluated.** Face angle and dynamic loft are defined at the *contact point*, not the face center. On a curved driver face (bulge and roll, Section 3.4) the local normal at a toe strike differs from the center normal by several degrees, so systems that measure face pose but not impact location are systematically biased on off-center hits.

2.1.1 Club path is reference-point dependent too — by about 3° on a driver

The reference-point issue is usually discussed only for club speed, but it applies with equal force to *direction*, and the magnitude is large enough to matter.

For any two points on the rigid clubhead,

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}, \quad \mathbf{r} = \mathbf{p}_P - \mathbf{p}_O, \quad (2.1)$$

so there is a *velocity field* across the head rather than a single path. A reported path is that field sampled at a chosen point. The two candidate points are far apart on a driver: the center of gravity sits roughly 25–50 mm behind the face, and the geometric center that radar tracks is typically within 6 mm of the CG [69].

The industry has split along the sensing modality. Radar systems report path at the geometric center, because that is the point the silhouette reconstruction can locate from behind. Optical systems with face fiducials report it at the *face center*, because that is where the markers are. TrackMan quantifies the resulting gap directly: for a driver, the CG path and the face-center path differ by **approximately 3°**, with the face-center path being the more out-to-in of the two, and the discrepancy shrinks for shorter clubs as the CG-to-face distance falls [62].

Two rotations drive it, both pushing the same way horizontally: the swing arc curves leftward through impact, so a point displaced forward along the arc has its velocity rotated further around it; and face closure about the shaft axis swings a point ahead of the rotation center leftward. The same geometry tilts the face-center velocity slightly *upward* relative to the CG — a shallower attack angle — because the face center sits ahead of the CG on an arc that is turning upward through the bottom.

Key point

Because the offset is $\boldsymbol{\omega} \times \mathbf{r}$, it is not a fixed constant: it scales with closure rate and arc tightness. Two players with identical reported path but different release rates do not have the same face-center path. And note that TrackMan’s reported face-to-path is already a *hybrid* — face orientation evaluated at the contact point minus velocity direction evaluated at the geometric center — so it is not a physically clean angle under either convention.

Design implication

This is the sharpest available argument for the twist formulation of Chapter E. Estimating $\xi = (\boldsymbol{\omega}, \mathbf{v}_O)$ makes path at any point a projection of one fitted object, so both conventions — and the rotation rate that separates them — fall out of the same estimate. An implementation should report path at a *declared* reference point, state which one, and expose $\|\boldsymbol{\omega}\|$ alongside it so the user can see how much the choice is worth on that swing. Reporting a bare path number without its reference point is reporting a quantity that is $\sim 3^\circ$ ambiguous on a driver.

2.2 Ball-Launch Parameters

Table 2.2: Ball-Launch Parameters (Defined Immediately After Separation).

Parameter	Definition
Ball speed	Speed of the ball’s center of gravity at separation
Smash factor	Ball speed $\div$ club speed
Launch angle	Vertical takeoff angle vs. the horizon
Launch direction	Horizontal takeoff angle vs. the target line
Spin rate	Rotation rate about the (single) spin axis, in rpm
Spin axis	Tilt of the rotation axis relative to the horizon; $-$ = tilted left $\Rightarrow$ draw for a right-hander
Carry / side / total	Trajectory descriptors; carry is measured to the point at launch elevation
Apex, landing angle	Peak height; descent angle at landing

A golf ball in flight has exactly one rotation vector $\boldsymbol{\omega}$. “Backspin” and “sidespin” are components of that vector, not separate spins. Systems that report sidespin (SkyTrak-style) and systems that report spin axis (TrackMan-style) are related by

$$S_{\text{side}} = S \sin \theta_{\text{axis}}, \quad S_{\text{back}} = S \cos \theta_{\text{axis}}, \quad \theta_{\text{axis}} = \text{atan2}(S_{\text{side}}, S_{\text{back}}). \quad (2.2)$$

A useful rule of thumb from TrackMan’s data: 1° of spin-axis tilt produces roughly 0.7% of carry as side-curve (about 0.7 yd per 100 yd) [45].

2.3 The Directness Hierarchy

For any launch monitor, each reported parameter falls somewhere on a hierarchy from directly measured to purely modeled. Anticipating the detailed treatment in Chapters 4 and 5, the hierarchy for the two sensing families is summarized in Table 2.3. This table is the skeleton of this whole review.

Table 2.3: Measured vs. derived, by architecture. “M” = measured directly, “D” = derived through a physical model, “E” = estimated/model-fit, “–” = not available.

Parameter	Radar (outdoor)	Radar (indoor)	Camera (photometric)	Hybrid
Ball speed	M	M	M	M
Launch angles	M	M	M	M
Spin rate	M ^a	M/E ^a	M ^b	M
Spin axis	D ^c	E	M ^b	M
Carry	M	E ^d	E ^d	E ^d
Club speed	M ^e	M ^e	M ^f	M
Club path / attack	M	M	M ^f	M
Face angle	D ^g	D ^g	M ^f	M
Dynamic loft	D ^g	D ^g	M ^f	M
Impact location	–	–	M ^f	M ^h

^a Doppler harmonic sidebands; requires surface asymmetry and sufficient flight (Section 4.4.2). ^b Dimple-pattern registration (Section 5.2.2). ^c Inverted from trajectory curvature via the Magnus constraint (Section 4.4.3). ^d Trajectory model integration from measured launch (Chapter 8). ^e Reference-point dependent; silhouette reconstruction on TrackMan. ^f Requires face fiducials (Foresight/Garmin R50) or overhead geometry (Uneekor/ProTee). ^g D-plane inversion from ball launch + path on radar-only units; optically assisted on OERT-class hardware (Section 4.5.2). ^h Markerless via camera on TrackMan 4/iO.

Chapter 3

Impact Physics: From Club Delivery to Ball Launch

Every launch monitor embeds a model of the club–ball collision. Camera systems use it *forward* (sanity-checking and filling gaps); radar systems use it *inverse* (recovering face angle and dynamic loft from measured ball launch). This chapter develops that model.

3.1 The D-Plane

Jorgensen [28] observed that at impact two unit vectors determine the collision geometry for a center strike:

- $\hat{n}$, the **face normal**, built from face angle φ_f and dynamic loft λ_d ;
- $\hat{p}$, the **club-head velocity direction**, built from club path φ_p and attack angle α .

These two vectors span a wedge-shaped plane — the *D-plane* (“descriptive plane”). Its two governing consequences are:

1. **Launch direction lies in the D-plane**, between $\hat{n}$ and $\hat{p}$, much closer to the face normal.
2. **The spin axis is normal to the D-plane:**

$$\hat{\omega} \propto \hat{p} \times \hat{n}. \quad (3.1)$$

The ball therefore curves *away from the path, around the face* — the “new ball flight laws.”

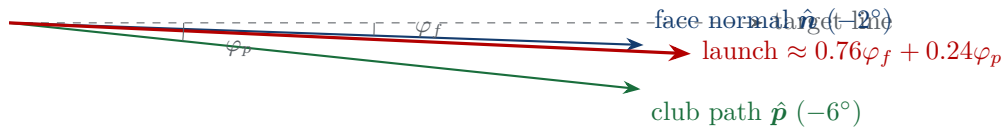


Figure 3.1: Horizontal D-plane geometry (top view, right-handed golfer, out-to-in “fade” delivery). The ball launches close to the face direction; the face-to-path difference tilts the spin axis and curves the ball away from the path.

3.1.1 Face/Path Weighting of Launch Direction

The filtered player data of Wood et al. [83] give the following study-specific horizontal weights in an approximate model of launch direction between face angle and club path:

$$\varphi_{\text{launch}} \approx w_f \varphi_f + (1 - w_f) \varphi_p, \quad w_f \approx \begin{cases} 0.76 \pm 0.08 & \text{driver} \\ 0.69 & \text{7-iron} \\ 0.61 & \text{wedge.} \end{cases} \quad (3.2)$$

Caution

The values above are the *horizontal* weights measured by Wood et al. [83]. Their player data contain 1,575 shots before filtering, with 157 golfers testing driver/7-iron and ten also testing a wedge. Club motion was measured optically at 720 fps and ball launch with a GC2. The horizontal means use strikes within 0.25 in of face centre in each axis and at least 1.5° face-to-path separation: 89 driver, 172 iron and 28 wedge shots remained. A separate 15-shot PING Man robot test returned 0.63 for the 7-iron.

The familiar 0.85/0.15 figure comes from TrackMan's *Ten Fundamentals*, which gives driver rules of thumb in both planes (and 75% for irons). Wood's driver means are 0.83 ± 0.08 vertically and 0.76 ± 0.08 horizontally, reported with 95% confidence intervals. The difference from 0.85 is a reason to retain the model's conditions, not a universal calibration correction. These are estimates from filtered samples, not individual-shot tolerance bands; protocol and measurand differences preclude treating the comparison as a definitive test of every application of the vendor rule.

This matters directly for any radar-first system: inverting Eq. (3.2) to recover face angle amplifies any bias in measured launch direction by $1/w_f$ when path and weight are fixed. At $w_f = 0.76$ the factor is 1.32, versus 1.18 at $w_f = 0.85$: about 12% greater sensitivity, not a factor of two. Path error and uncertainty in the weight contribute separately, so this derivative is not a complete uncertainty budget for inferred face angle.

The weight is loft-dependent: more loft means a more oblique impact, hence a larger tangential momentum component along the path direction. Peer-reviewed impact modeling with normal COR plus tangential compliance reproduces the loft dependence from first principles [59, 83].

The *mechanism* is contested. TrackMan attributes the driver's higher weight to a smoother, lower-friction titanium face; PING rejects this, showing the loft dependence persists at constant μ and that below $\sim 20^\circ$ of incidence *lower* friction moves launch *closer* to the path—the opposite of the friction hypothesis [59]. An implementation that fits w_f empirically should record which explanation its calibration assumes.

Tutelman [72] gives a closed-form version calibrated to TrackMan data. With face-to-path angle A and dynamic loft L , define the total obliqueness Φ (which is precisely the spin loft):

$$\cos \Phi = \cos A \cos L, \quad f(\Phi) = 0.96 - 0.0071 \Phi \quad (\Phi \text{ in degrees}), \quad (3.3)$$

and the departure angles relative to the club path are

$$\text{DA}_{\text{vert}} = L f(\Phi), \quad \text{DA}_{\text{horiz}} = A f(\Phi), \quad (3.4)$$

with launch angles relative to the target obtained by adding attack angle and path respectively. Note $f(\Phi) \approx 0.85$ at $\Phi \approx 15^\circ$ (driver) and ≈ 0.75 at $\Phi \approx 30^\circ$ (short iron): one friction-calibrated function reproduces both rules of thumb.

3.1.2 Spin-Axis Tilt

For a center strike the spin-axis tilt follows from the D-plane geometry:

$$\tan \theta_{\text{axis}} \approx \frac{\tan(\text{face-to-path})}{\tan(\text{spin loft})}, \quad (3.5)$$

so 1° of face-to-path tilts the axis $\approx 4^\circ$ with a driver (spin loft $\sim 12\text{--}15^\circ$) but only $\approx 2^\circ$ with a mid-iron (spin loft $\sim 25\text{--}30^\circ$) [45]. Tuxen's shot-shaping rule for a ball that curves back to the target line: $\theta_{\text{axis}} \approx -2.5 \times$ the horizontal launch angle.

Design implication

A radar-first system typically measures launch direction and club path. Equation (3.2) then yields face angle by inversion: $\varphi_f = (\varphi_{\text{launch}} - (1 - w_f)\varphi_p)/w_f$. Because $1/w_f$ depends on the assumed weight, launch-direction bias alone is amplified by about 18% at $w_f = 0.85$ or 32% at $w_f = 0.76$, with path and weight fixed. Alignment calibration matters, but its priority relative to path error, weight uncertainty and model discrepancy requires a complete uncertainty budget.

3.2 Ball Speed and Smash Factor

The normal-direction collision is well modeled as a two-body impact with coefficient of restitution e [6, 44]:

$$v_{\text{ball}} = v_{\text{club}} \frac{1 + e}{1 + m/M} \cos \Phi (1 - 0.14 x_{\text{miss}}), \quad (3.6)$$

with $e \approx 0.83$ at the USGA driver limit (falling with loft and impact speed), ball mass $m = 45.93$ g, head mass $M \approx 200$ g (driver), $\cos \Phi$ the obliqueness loss, and the last factor an empirical off-center loss with x_{miss} in inches [71]. The prefactor $(1 + e)/(1 + m/M) \approx 1.49$ sets the theoretical driver smash ceiling; smash factor falls with loft:

Effective loft	0°	10°	20°	30°
Max smash factor	1.488	1.465	1.398	1.288

Key point

This table is a built-in sanity check: a measured smash factor above the loft-appropriate ceiling indicates a measurement error — most commonly a radar unit reporting toe speed or shaft speed instead of geometric-center club speed. An implementation should flag physically impossible smash values instead of displaying them.

3.3 Spin Generation

Friction during the oblique compression converts tangential surface speed $v_{\text{club}} \sin \Phi$ into rotation. Real impacts exhibit tangential compliance (the contact patch sticks and stretches like a shear spring) [59, 7], but a practical engineering fit calibrated to TrackMan data [72] is simply

$$S_{\text{total}} [\text{rpm}] \approx 160 v_{\text{club}} [\text{mph}] \sin \Phi, \quad (3.7)$$

with the back/side split following from the D-plane direction ratio $d = \tan A / \tan L$: $S_{\text{back}} = S / \sqrt{1 + d^2}$, $S_{\text{side}} = d S_{\text{back}}$. Cross-checks: ~ 200 – 300 rpm per degree of spin loft for a driver; the “iron loft $\times 200$ ” rule matches PGA Tour averages. Friction saturates near spin lofts of 45 – 50° (the “spin-loft cliff”), and wet or grass-contaminated contact reduces spin at fixed spin loft [66].

3.4 Gear Effect and Impact Location

An impact at horizontal offset x from the CG line torques the head about its CG; the recoiling face “gears” the ball the opposite way. The angular-impulse model [70] gives head rotation

$\omega_{\text{head}} = x m v_{\text{ball}}/I_h$ and gear spin

$$s \text{ [rpm]} = 58,830 \frac{v_{\text{ball}}[\text{mph}] C[\text{in}] x[\text{in}]}{I_h[\text{g cm}^2]} \approx 16.4 v_{\text{ball}}[\text{mph}] x[\text{in}] \times 100, \quad (3.8)$$

where C is CG depth behind the face (32–47 mm on measured OEM drivers) and $I_h = 4000\text{--}5800 \text{ g cm}^2$; the ratio I_h/C is nearly constant across modern drivers, which is why the simplified form works to $\pm 2.5\%$. A toe strike opens the head and imparts *hook* gear spin; a heel strike, slice spin. Vertical gear effect (high-face strikes reduce backspin, low-face strikes increase it) follows the same equation with the vertical CG offset and roll radius.

Driver faces are built curved to exploit this: **bulge** (horizontal radius, typically 12 in) starts a toe miss right so the gear-effect hook curves it back. Worked example [70]: a 1 in toe miss at 150 mph ball speed gets +1354 rpm of bulge-induced slice spin against -2192 rpm of gear hook spin — net 838 rpm hook and ~ 10 yd left, versus ~ 61 yd offline for a flat face. Sensitivity is extreme: a strike one dimple width (0.14 in) off-center tilts the spin axis $\sim 6^\circ$ on a driver ($\sim 2^\circ$ on a 6-iron) [45]. Irons have shallow CG depth, hence weak gear effect and flat faces.

Design implication

Gear effect is why *impact location* is the most valuable club parameter a launch monitor can add after path/face: without it, a pure D-plane inversion misattributes gear-effect spin-axis tilt to face-to-path. For a radar-only system this is a fundamental, quantifiable error source on off-center driver strikes (up to several degrees of apparent face-to-path); Equation (3.8) bounds it, and an optical impact-location add-on (Chapter 10) removes it.

3.5 Vertical Plane and the Low-Point Geometry

The identical geometry applies vertically: launch angle sits 75–85% of the way from attack angle toward dynamic loft, and spin loft sets spin magnitude via Equation (3.7). One non-obvious coupling: for a descending strike on an inclined swing plane, the instantaneous path points in-to-out even when the swing direction is square — e.g. hitting 5° down on a 60° plane yields $\approx 2.9^\circ$ of in-to-out path. Radar systems that measure swing plane and attack angle use this relation (path = f (swing direction, swing plane, attack angle)) as a consistency constraint among the measured club parameters.

Chapter 4

Doppler Radar Systems

4.1 Continuous-Wave Doppler Fundamentals

A CW radar transmits a carrier at frequency f_c (wavelength λ) and receives echoes shifted by the Doppler effect. For a target with radial velocity v_r ,

$$f_d = \frac{2v_r}{\lambda} = \frac{2f_c v_r}{c}. \quad (4.1)$$

At the 24.125 GHz K-band center frequency used by the OPS243-A (and the Garmin R10, Full Swing KIT, original Mevo), each 1 mph of radial speed produces ≈ 71.7 Hz of shift; at X-band (~ 10.5 GHz, TrackMan’s long-range subsystem and the Mevo+) the constant is ≈ 31.3 Hz/mph. A spinning, translating golf ball is not a point target: every surface patch has its own radial velocity, so the return is a velocity *spectrum* whose structure carries the spin information (Section 4.4.2).

Band choice is a range-vs-resolution trade. X-band supports long-range full-flight tracking (TrackMan tracks the entire ~ 6 s flight); 24 GHz gives finer velocity resolution per unit observation time and compact low-power hardware, but consumer 24 GHz units track only ~ 30 yd of flight [9, 19, 37].

4.2 From Velocity Sensor to 3D Tracker: Phase Interferometry

A single-channel Doppler radar measures radial speed only. Every serious launch monitor adds *multiple receive antennas* and measures the *phase difference* of the return across receiver pairs (phase-comparison monopulse / interferometry). A wavefront arriving from direction $\hat{\mathbf{u}}$ reaches two antennas separated by baseline $\mathbf{d}$ with time delay $\tau = (\mathbf{d} \cdot \hat{\mathbf{u}})/c$, observed after mixing as a phase difference

$$\Delta\varphi = 2\pi f_c \tau \pmod{2\pi} \implies u = \frac{\lambda \Delta\varphi}{2\pi d}, \quad (4.2)$$

where u is the direction cosine along the baseline. Two orthogonal baselines give azimuth and elevation; the 2π ambiguities are resolved with more than three antennas at staggered spacings (TrackMan US 9,958,527 [55]). Combined with range (from multi-frequency CW phase differences or an FMCW chirp) and range rate (Doppler), the radar produces a full 3D track $(r, \text{az}, \text{el}, \dot{r})$ per epoch. Tuxen’s spin-axis patent describes exactly this “phase–phase monopulse comparison . . . or interferometry” across three receivers [74]; the Garmin R10 uses a three-receiver 24 GHz array the same way [37]; FlightScope describes its aperture as a phased antenna array [58].

Design implication

A minimal radar architecture — one CW Doppler module for speed plus two angle-radar modules for vertical and horizontal angles — is a physically separated version of what commercial units do on one board: the K-LD7’s RADC mode exposes raw ADC data from its two receive patches, and per-bin phase comparison Equation (4.2) yields the angle of each detection. The commercial

lesson is that angle accuracy is limited by baseline length, SNR, and mutual calibration; a rigid mechanical mount and a one-time angular calibration against surveyed targets matter more than any software refinement.

4.3 CW + FMCW Hybrids

Full Swing’s patent US 11,311,789 (published as US 2020/0147470) [8] is the clearest public description of a hybrid launch-monitor radar: two ~ 24 GHz transmitters, time-division multiplexed, one CW and one linear-FMCW. The CW channel provides club speed, ball speed, spin, and launch-window angles from Doppler; the FMCW chirp provides direct range to 250 m+, carry, and side displacement from beat-frequency and phase measurements. The antenna layout is deliberately *non-uniform* — four receive antennas symmetric about the axis, two transmit antennas placed asymmetrically — to enrich the phase/frequency structure of the receive signal. FlightScope’s X3 similarly advertises “multi-frequency radar with direct distance measurement” [58].

4.4 Ball Measurement

4.4.1 Speed and Launch Angles

Ball speed is read from the dominant Doppler line immediately after impact; launch angles come from the interferometric track over the first meters of flight. These are the most directly measured quantities in the whole parameter set and show the best inter-device agreement in every independent study (Chapter 9). Because the radar sits behind (or above) the ball, the measured radial speed underestimates the true speed by the cosine of the aspect angle; commercial units correct this using the tracked geometry, and the correction is largest for high launch angles and close radar placement.

4.4.2 Spin Rate: The Harmonic-Sideband Method

The foundational method is Tuxen’s patent US 8,845,442 [73] (EP 1 698 380). A golf ball’s dimples, paint, logo, seam, and core asymmetry make its radar cross-section vary periodically as it rotates. A surface feature at radius r contributes a Doppler term

$$f_{d,\text{feature}}(t) = \frac{2}{\lambda}(v_r - r\omega \sin \omega t), \quad (4.3)$$

i.e. frequency modulation of the return at the spin frequency $\omega/2\pi$ with deviation $(2/\lambda)r\omega$. The received spectrum therefore shows the main translational Doppler line flanked by **equally spaced harmonic sidebands** at

$$f = f_d \pm n f_{\text{spin}}, \quad n = 1, 2, 3, \dots \implies S [\text{rpm}] = 60 \Delta f_{\text{sideband}}. \quad (4.4)$$

The patented processing chain: (1) track the central velocity trace through the short-time Fourier transform; (2) detect symmetric sideband peaks; (3) track them over time as spectral traces; (4) *qualify* harmonics by verifying constant equal spacing across consecutive FFT frames; (5) solve the harmonic-number assignment; (6) divide trace offset by harmonic number [73, 64]. In practice a cepstrum or harmonic-product-spectrum estimator finds the comb spacing robustly.

Two practical corollaries: sideband SNR depends on ball-surface asymmetry (pristine two-piece range balls read weakly — the origin of “marked-ball” modes and Titleist RCT balls with embedded metal tags); and the comb yields spin *rate* only — the axis must come from elsewhere.

4.4.3 Spin Axis: Trajectory Inversion

The same patent discloses the standard radar route to spin axis. From the differentiated 3D track, total acceleration $\mathbf{A}$ decomposes as $\mathbf{A} = \mathbf{g} + \mathbf{D} + \mathbf{L}$ with drag $\mathbf{D}$ antiparallel to the airspeed vector and Magnus lift $\mathbf{L}$ perpendicular to both the spin axis and the velocity. Projecting out gravity and drag,

$$\mathbf{D} = \left[\frac{(\mathbf{A} - \mathbf{g}) \cdot \mathbf{v}_a}{\|\mathbf{v}_a\|^2} \right] \mathbf{v}_a, \quad \mathbf{L} = \mathbf{A} - \mathbf{g} - \mathbf{D}, \quad (4.5)$$

and the orthogonality constraint $\mathbf{L} \cdot \hat{\boldsymbol{\omega}} = 0$, stacked over many trajectory points, gives an overdetermined linear system for the spin-axis direction $\hat{\boldsymbol{\omega}}$ [73]. This works well outdoors with seconds of observed flight; indoors, with 2–6 m of flight before the screen, the curvature signal is tiny and the axis becomes an estimate. FlightScope patented a direct alternative — time delays between vertically and horizontally separated receiver pairs yield the axis angle $\Phi = \arctan[S_H T_H / (S_V T_V)]$ [26] — precisely to recover the axis at short range.

4.4.4 Flight: Tracked Outdoors, Modeled Indoors

Outdoors, TrackMan-class units report *measured* carry, apex, and landing angle from the full track. Indoors every radar unit observes only the pre-screen segment (TrackMan 4 requires ≥ 4.7 m of throw; FlightScope claims 8 ft suffices with Fusion Tracking) and extrapolates with an aerodynamic model anchored to the vendor’s measured-flight database [63, 23]. The Garmin R10 illustrates the consumer floor: it requires ≥ 20 m of visible flight and ball speed above ~ 90 mph to measure spin at all; otherwise spin is estimated by a fitted model from the measured inputs [36].

4.5 Club Measurement

4.5.1 What the Radar Tracks on the Head

The clubhead is a large, complex reflector whose parts move at different speeds (toe faster than heel by up to ~ 7 mph on a driver). TrackMan defines club speed at the head’s *geometric center* and reconstructs that point by “directly measuring the 3D silhouette of the club head” [69]; naive processors lock onto the strongest or fastest return and systematically over-report. Attack angle and club path are the vertical and horizontal direction of the geometric center’s velocity at impact; swing plane, swing direction, and low point are properties of the center’s trajectory arc through the hitting zone.

4.5.2 Face Angle and Dynamic Loft

Radar cannot see face orientation. On radar-only systems, face angle and dynamic loft are obtained by *inverting the D-plane collision model* (Section 3.1.1): given measured launch direction, club path, and the loft-dependent weighting w_f ,

$$\varphi_f = \frac{\varphi_{\text{launch}} - (1 - w_f) \varphi_p}{w_f}, \quad (4.6)$$

and analogously dynamic loft from launch angle and attack angle. TrackMan has confirmed these are “derived numbers from direct measurements and a collision model,” validated by robot testing [24]. On TrackMan 4/iO with OERT (Section 4.6), the company’s position is that club delivery including face angle at the contact point is measured via synchronized radar+camera silhouette tracking

rather than back-computed — though the exact split remains proprietary. Consumer units are explicit about the hierarchy: Garmin specifies launch direction at $\pm 1^\circ$ (measured) but face angle at $\pm 2^\circ$ and classifies it as calculated [36].

Key point

The reported “face angle” on a radar launch monitor is, to first order, a rescaled launch direction. It inherits launch-direction bias amplified by $1/w_f \approx 1.15$, plus a gear-effect error on off-center strikes that the model cannot see (Section 3.4). This is not a defect of any particular product but a structural property of the modality.

4.5.3 Known Limitations

Independent benchmarking consistently finds club parameters far less accurate than ball parameters for radar (and for camera systems without face markers): the Leach et al. criterion study is the key reference [31]. Irons are harder than drivers (shaft and hosel returns, turf strike clutter, lower speeds); TrackMan’s advertised “>90% club-data pickup rate” with OERT is itself an acknowledgment that pickup is not universal [68]. Alignment error translates directly into correlated path/face bias on all units.

4.6 Optically Enhanced Radar: OERT and Fusion Tracking

TrackMan 4 fuses its dual radar (X-band long-range ball subsystem + 24 GHz high-resolution club/impact subsystem, receivers sampled at 40 ksps) with a built-in camera synchronized in time and space: the camera contributes exact pre-impact ball position, markerless *impact location* on the face, and “4D silhouette” clubhead tracking between radar epochs [68, 9]. The indoor-only TrackMan iO packs a 24 GHz radar with dual cameras — one at up to 4,600 fps for club and ball, one for alignment — under 810–850 nm IR illumination [63]. FlightScope’s patented equivalent is Fusion Tracking (US 10,338,209 [25]): the radar’s predicted track steers image search windows, the camera’s precise angular fixes correct the radar’s, and inter-sensor offsets are removed by error minimization. Rapsodo’s MLM2PRO pairs radar with 240 fps cameras that read the printed dot pattern on RPT/RCT balls for measured spin rate and axis [38]; Full Swing’s KIT feeds a 4K camera into an ML pipeline alongside its CW+FMCW radar [14].

Key point

Every major radar vendor has independently concluded that pure Doppler is insufficient at short range — for spin axis, impact location, and club-face data — and has added optical sensing or instrumented balls. This is the strongest architectural signal in the market for a radar-first design’s roadmap.

Chapter 5

Photometric (Camera) Systems

5.1 Imaging Architectures

Photometric launch monitors capture a short burst of high-speed, IR-strobed stereo images over a small capture volume beside (or above) the tee — roughly the first 30 cm of ball flight — and reconstruct ball and club kinematics photogrammetrically. The commercial family tree:

- **Foresight Sports** defined the reference floor-unit design. The GC2 (2010) used two cameras (“stereoscopic”), with club data via the add-on HMT module (two more cameras). The GC3 / Bushnell Launch Pro (a Foresight-built rebadge) use three cameras; the GCQuad and QuadMAX use four cameras at the corners of the sensor window, capturing on the order of 200 images of ball and club per shot at burst rates quoted up to 10,000 fps; the ceiling-mounted GCHawk points the same quadrascopic package downward [11, 47]. Onboard IR object tracking detects the teed ball and the inbound clubhead and triggers the burst; intrinsics/extrinsics are factory-calibrated (“no calibration needed”) [11].
- **Uneekor** mounts overhead: EYE XO/XO2 look straight down from ~ 3 m with two (XO2: three) high-speed IR cameras at 3,000+ fps [75]. The overhead geometry sees the clubhead and face region directly, enabling sticker-free club data and impact-location video replay (Club Optix). The portable EYE MINI reverts to a side view and therefore requires club stickers.
- **ProTee VX**: overhead, two synchronized high-speed cameras with IR illumination and an ML shot-analysis pipeline; sticker-free club and ball data including vertical/horizontal impact point [48].
- **Garmin Approach R50**: three high-speed cameras arranged horizontally in a floor unit; purely optical ball and club measurement, with clubhead data requiring reflective fiducials [15].
- **SkyTrak / SkyTrak+**: the original SkyTrak was photometric and ball-only; the SkyTrak+ added a dual Doppler radar module specifically for club data while the cameras handle ball launch and spin [53].

5.2 Ball Measurement

5.2.1 Photogrammetric Position and Velocity

With factory-known intrinsics K_i and extrinsics $[R_i|t_i]$, each synchronized image set yields a 3D ball-center fix by triangulation: back-project the matched detections as rays and solve the least-squares intersection (or the DLT system $\tilde{\mathbf{x}} \times (P\mathbf{X}) = 0$), refining by reprojection error. Ball speed is the displacement between timestamped fixes; launch angle and direction are the components of the initial velocity vector in the calibrated target frame. Two error sources dominate: depth error

grows quadratically with distance,

$$\sigma_Z = \frac{Z^2 \sigma_{px}}{f B}, \quad (5.1)$$

for baseline B and focal length f — which is why photometric units keep the capture volume small and also exploit the known ball diameter (21.34 mm radius) as an auxiliary range cue; and center-finding bias — fitting the projected *conic* outline rather than taking the blob centroid avoids a systematic offset of up to RZ/f pixels.

5.2.2 Spin: Dimple-Pattern Registration

The flagship photometric capability is markerless 3D spin. Successive frames of the ball are registered on the sphere: the estimator searches the rotation $R \in SO(3)$ that best maps the back-projected surface texture (the dimple pattern, plus any logo or blemish) of frame k onto frame $k+1$ — Foresight brands this *spherical correlation*; Uneekor, *Dimple Optix* [10, 77]. The recovered R gives everything at once:

$$\hat{\omega} = \text{eig}_1(R), \quad \theta = \arccos\left(\frac{\text{tr } R - 1}{2}\right), \quad \omega = \theta/\Delta t, \quad (5.2)$$

i.e. the spin axis is the eigenvector of R with unit eigenvalue and the spin rate follows from the per-frame rotation angle. Reported accuracy is 1–3% of true spin, independent of ball speed. The known failure mode is rotational aliasing at high spin and low frame rate (the $n \cdot 2\pi$ ambiguity), resolved by multi-hypothesis tracking or a trajectory-consistency prior. Where resolution cannot support dimple registration, systems fall back to *marked balls*: Uneekor QED and Rapsodo (RPT balls) track printed high-contrast features, giving closed-form rotation from ≥ 3 correspondences via the Kabsch/Horn algorithm.

5.3 Club Measurement

5.3.1 Fiducial Markers: The Foresight Approach

Foresight floor units require retroreflective **fiducial dots** on the clubface. The dot count maps directly to the data tier [12, 13]: one dot suffices for club speed, path, and attack angle (tracking a single 3D point through the capture volume); *four dots* — placed on the vertical centerline, equidistant from the horizontal centerline — define the face plane and enable face angle, dynamic loft, lie, closure rate ($^\circ/\text{s}$), and **impact location**. Mechanically: the IR strobe makes the dots bloom in every frame; triangulating the dot constellation across cameras and frames yields the 6-DOF pose of the face through impact. Club speed is the pose translation rate; path and attack angle are the velocity direction; face angle and dynamic loft are the face-plane orientation at contact; impact location is the measured ball position expressed in the face frame. The GC3/Launch Pro tier delivers speed, path, attack, face angle, and impact via the same stickers but omits dynamic loft, lie, and closure rate — these remain GCQuad/QuadMAX exclusives [47].

5.3.2 Sticker-Free Overhead Tracking

Uneekor’s overhead cameras segment the clubhead silhouette directly — no markers — because the top-down view presents the head against the mat with controlled IR illumination; the systems also record actual high-speed impact video (face-on via Club Optix), from which impact location is read [75]. ProTee VX claims the same sticker-free club set (speed, path, face, attack, dynamic loft, lie, vertical and horizontal impact point) from its overhead ML pipeline [48]. The trade-off is installation: overhead units need $\sim 3\text{ m}$ ceilings, rigid mounting, and a floor-chart calibration.

5.3.3 Calibration

Factory-rigid multi-camera rigs ship calibrated; the user aligns only the target line. Overhead systems require field calibration: Uneekor’s procedure places a printed calibration chart on the mat, levels it, squares it to the screen, and aligns its crosshair to overlays on the live camera feeds — establishing the ground plane, hitting zone, and target-line extrinsics [4]. This is the pattern a DIY overhead build should copy.

5.4 The Indoor Argument, and What Cameras Cannot Do

Photometric systems measure actual launch conditions — including 3D spin — in the first foot of flight; nothing is lost when the ball hits a screen 2.5 m away. Radar indoors must infer from a truncated flight. This is the structural reason camera units dominate indoor simulation while radar dominates outdoor practice: each modality measures what the other models [21]. The camera’s blind spots are downrange (all flight is simulated from launch) and, for floor units, the club face itself without stickers — the same face-visibility problem radar has, solved with markers instead of models.

5.5 DIY and Open-Source Photometric Systems

PiTrac [46] is the flagship open-source photometric build and the most relevant open-source precedent. Instead of multi-kfps cameras it uses *strobed multi-exposure capture*: an IR LED array pulses several times within one long exposure of a ~\$50 Raspberry Pi Global Shutter camera, freezing multiple ball images in a single frame. One camera watches the teed ball and detects launch; the second captures the strobed flight images; a custom PCB sequences trigger and strobe. Processing (C++/OpenCV): Hough-circle detection locates ball images; strobe-timestamped positions give speed and launch angles; 3D spin on all three axes is solved by searching candidate rotations that best register the dimple imagery between exposures — Gabor-filter dimple enhancement is documented in the adjacent patent literature [60]. Total BOM ≈\$250–300. The architecture shares multi-exposure imaging and surface-feature registration principles with the disclosures discussed in Section 7.2; that comparison establishes neither a direct product lineage nor rights clearance.

Design implication

PiTrac provides an open implementation of strobed optical capture and image-based motion estimation using commodity hardware. Its architecture can inform a radar/optical prototype, but a parts list or successful spin fit is not a measurement-error specification. Ball-spin capture and club-face impact imaging require their own visibility, calibration and validation checks; adding a camera does not automatically establish both capabilities.

Chapter 6

Commercial System Survey

Table 6.1 summarizes the sensing architecture of the major systems; the notes that follow give the technically salient details per device, with sources in the bibliography.

Table 6.1: Launch monitor architecture survey (2026). “Club source” distinguishes directly measured face data (optical) from D-plane-inverted or model-estimated face data.

System	Primary sensing	Spin method	Club data source	Impact location
TrackMan 4	Dual radar (X-band + 24 GHz) + camera (OERT)	Doppler sidebands	Radar silhouette + camera	Yes (markerless, OERT)
TrackMan iO	24 GHz radar + 4,600 fps camera	Camera + radar	Radar + camera	Yes (markerless)
FlightScope X3	Phased-array multi-freq. radar + Fusion cameras	Doppler (dielectric-lens)	Radar; face derived	No
FlightScope Mevo+	X-band 10.5 GHz phased array (+ Pro cameras)	Doppler; stickers indoors	Radar; face derived	No
Garmin R10	24 GHz, 3-receiver CW	Doppler (≥ 20 m flight) or model	Radar; face calculated	No
Garmin R50	3 high-speed cameras	Dimple imaging	Optical (fiducials)	Yes
Full Swing KIT	24 GHz CW+FMCW + 4K camera ML	Doppler + ML	Radar + ML	No
Rapsodo MLM2PRO	Radar + 2×240 fps cameras	Marked ball (RPT/RCT)	Radar; face derived	Video only
Foresight GCQuad / QuadMAX	4 cameras, IR strobe	Spherical correlation	Optical (4 fiducials)	Yes (measured)
GC3 / Launch Pro	3 cameras, IR strobe	Spherical correlation	Optical (fiducials; add-on)	Yes
GCHawk	4 cameras, ceiling	Spherical correlation	Optical	Yes
Uneekor EYE XO2	3 overhead IR cameras, 3,000+ fps	Dimple Optix	Optical, sticker-free	Yes + video
Uneekor EYE MINI	2 cameras, portable	Dimple Optix	Optical (stickers)	Yes
ProTee VX	2 overhead cameras + ML	Dimple imaging	Optical, sticker-free	Yes
SkyTrak+ / ST MAX	Cameras (ball) + dual radar (club)	Photometric	Radar; face derived	No
OpenFlight (open source)	OPS243-A 24 GHz + 2×K-LD7 + sound trigger	Doppler I/Q buffer (~ 50 – 60% detect)	Radar; no face data yet	No
PiTrac (DIY)	2 Pi GS cameras + IR strobe	Dimple registration	None yet	No

6.1 Radar-First Systems

TrackMan 4 / iO. The reference radar architecture: two synchronized radar subsystems (long-range X-band at the corners for full ball flight; higher-frequency 24 GHz at the center for club and impact), receivers sampled at 40 ksp/s to pin the impact instant, fused with an OERT camera [9, 68, 63]. Reports 27+ parameters including swing plane/direction, low point, dynamic lie, and D-plane-derived face data; requires ≥ 4.7 m of indoor throw. The iO is the indoor-optimized repackaging: 24 GHz radar, 4,600 fps club/ball camera, IR illumination, no minimum-space requirement.

FlightScope X3 / Mevo+. Phased-array Doppler (X3: multi-frequency with direct ranging; Mevo+: X-band 10.5–10.55 GHz per FCC filings [19]) with Fusion Tracking cameras on the X3 and Mevo+ Pro. Spin measured via the patented dielectric-lens phase-demodulation route (Section 7.3); metallic stickers recommended for short indoor flights. FlightScope’s 2022 win against TrackMan at the German Federal Court of Justice, after losing the 2013 EP 1 698 380 case, bookends two decades of radar-spin litigation between the two [18].

Garmin Approach R10. The instructive budget case: a 24 GHz three-receiver CW radar that directly measures only the safe primitives — ball speed, launch angles, club speed, club path — and models the rest. Spin requires ≥ 20 m of flight and >90 mph ball speed (or an RCT marked ball indoors); face angle is explicitly a calculated value at $\pm 2^\circ$ [36, 37]. Garmin’s step-up R50 abandons radar entirely for three cameras — a telling modality switch for indoor accuracy.

Full Swing KIT. The patented CW+FMCW dual-mode 24 GHz radar (Section 4.3) with an ML vision assist from its 4K camera; 16 parameters; third-party testing places it within 1–2% of TrackMan/GCQuad in most conditions [14].

Rapsodo MLM2PRO. Radar for speed/launch plus two 240 fps cameras that read printed RPT/RCT ball markers for measured spin rate and axis (claimed within 1% of reference units) — the cheapest route to *measured* spin, at the cost of proprietary balls [38].

6.2 Camera-First Systems

Foresight GCQuad / QuadMAX / GCHawk. Quadrascopic IR-strobed capture (~ 200 images/shot); spherical-correlation spin; four-dot fiducial club measurement with closure rate and measured impact location (Section 5.3) [11, 12]. Robot testing shows the class-leading spin repeatability (Chapter 9).

GC3 / Bushnell Launch Pro. Identical triscopic hardware in two brands; club data is a paid add-on tier using the same stickers [47].

Uneekor EYE XO2 / EYE MINI / ProTee VX. Overhead sticker-free club measurement (XO2, VX) versus portable sticker-based (EYE MINI); Dimple Optix markerless spin; impact video replay [75, 48].

SkyTrak+ / ST MAX. The camera-to-radar convergence case: the original photometric ball-only SkyTrak gained a dual-radar club module and ML fusion in the SkyTrak+ [53].

6.3 Open-Source Systems

OpenFlight (radar-first: OPS243-A I/Q rolling buffer + sound trigger + K-LD7 interferometric angle radars) and **PiTrac** (camera-first: strobed multi-exposure global-shutter imaging) occupy the two ends of the same spectrum the commercial market spans, at roughly 1/10th the hardware cost. Neither yet measures club face data; both have clear, patent-informed paths to it (Chapter 10).

Chapter 7

Patent Landscape

Patent publications expose proposed measurement architectures that can be hard to reconstruct from product brochures. Their technical value lies in the signal being observed, the calibration and inference steps, and the conditions under which an output is identifiable. A disclosure, a granted claim, a commercial implementation and an accuracy experiment are four different kinds of evidence. This chapter compares selected disclosures; Chapter D provides a wider catalog of document identifiers.

Read an abstract as an overview, a description as a set of disclosed embodiments, and each claim as its own combination of limitations. An example camera count, exposure time or receiver arrangement in the specification is not automatically required by every claim. Conversely, a broad title such as “spin measurement” does not mean that a patent covers every method producing that output. Company groupings below aid navigation; they do not replace assignment records or establish which product implements a particular embodiment.

7.1 TrackMan A/S (Interactive Sports Games A/S) — Fredrik Tuxen

TrackMan’s virtual-marking notice contains 47 distinct US patent numbers at the October 1, 2026 check [65]. That dated list is a route into its disclosures, not an exhaustive search of applications, foreign publications or the company’s ownership history.

Spin Frequency and Trajectory-Based Axis. US8845442B2. The disclosure describes radar spectral spacings associated with spin and a separate trajectory-based approach to spin axis [73]. Claim 1 concerns estimating spin frequency from a frequency distance in the modulated radar signal; it should not be paraphrased as requiring all the axis-inversion steps in the description. Spectral structure still needs a defensible mapping from the observed periodicity to ball rotation. Trajectory-based axis inference also depends on the aerodynamic model and the information present in the observed flight segment (Section 4.4.2 and Equation (4.5)).

Target-Referenced Measurements. US8085188B2 / US9857459B2 / US10473778B2. US8085188B2 determines direction deviation relative to an image-designated target; US9857459B2 adds a co-rotated camera/radar arrangement in its first claim. US10473778B2 claim 1 instead correlates radar trajectory data with an image feature and displays the resulting trajectory information. Related coordinate transformations appear in US8912945B2 and US10690764B2: a launch position and image-identified reference define a target-relative frame. These operations change how an observation is related to the scene; they do not by themselves add an independent observation of projectile motion. Sensor and image coordinates must be registered before a target-relative angle has a physical meaning.

A Radar View of Spin Axis. US10850179B2. Claim 1 separates the Doppler spectrum into components, determines their angular positions using a non-collinear receiver arrangement and

identifies a *projection* of the spin axis onto a plane perpendicular to the radar’s line of sight [74]. That projection is not, by itself, an unrestricted three-dimensional axis measurement. Motion compensation, receiver geometry and the distinction between a projected axis and a full axis belong in any explanation of the method. US11446546B2 claim 1 likewise identifies a projected axis; US11938375B2 claim 1 identifies an axis component from phase differences for radar frequency components. The related publications US11446546B2 and US11938375B2 therefore require their own account of the output and receiver geometry.

Tracking, Calibration and Impact. US10989791B2 and US11828867B2 concern radar/imager tracking; US11619708B2 and US12517218B2 address inter-sensor calibration; US10953303B2 describes estimating club impact information from image features. These are distinct inference problems, even when a product presents their outputs together. The similarly titled radar/image documents US10596416B2, US11697046B2 and US12128275B2 identify Topgolf Sweden AB, not TrackMan, in their publication records. In US11619708B2 claim 1, the initial camera track already uses radar range together with image angles. Agreement between that track and the radar track is therefore not a comparison of two wholly independent measurements. US12517218B2 claim 1 selects corrections by minimizing quality-weighted three-dimensional track residuals. Improved correspondence is a calibration objective; agreement alone does not establish absolute accuracy against an independent reference.

Display and Attribution. Trajectory illustration appears in US10315093B2. Associating tracked balls with launch locations is described in US11086005B2. A convincing video overlay and a correct association with a player are separate requirements from accurate trajectory measurement.

Terrain-Constrained Position. US12186643B2 intersects a calibrated camera-to-ball ray with a three-dimensional terrain model and outputs that intersection as a ball position. The terrain constraint supplies depth information that the image alone lacks. For interpreting such an output, distinguish the ray–surface intersection from independently observed ball height: during the airborne part of a bounce, a ball above the surface need not coincide with that intersection. An output labeled three-dimensional therefore still needs an account of the geometric assumptions that determined it.

7.2 Foresight Sports, WAWGD and Wintriss Disclosures

The Wintriss/WAWGD documents provide examples of image-based launch measurement and integrated monitor operation. Uneekor’s product marking notice lists several of the same numbers, and the companies announced a license agreement in 2024 [76, 3]. That evidence should not be expanded into an assertion that every Foresight or Uneekor device implements every disclosed feature.

Surface Images. US7292711B2 / US7324663B2. The disclosures describe recovering ball motion from images of surface features, including dimples and non-precision marks [29]. US7292711B2 claim 1 includes compensation for sensor misalignment relative to a calibrated orientation. US7324663B2 claim 1 describes a search over rotations, scaling and correlation scores, together with coordinate and alignment calculations. Here, “markerless” means that prescribed fiducials need not be applied; it does not mean that the measurement needs no resolvable texture, lighting control or calibration. Correlation quality and rotation ambiguity must be evaluated on the actual images.

Trigger and Placement. US7497780B2 / US7641565B2. The former integrates trigger-zone monitoring, capture, parameter calculation and output. The latter detects ball placement using images and physical attributes. Reliable triggering determines which frames reach a spin estimator; it is part of the measurement chain, rather than merely a display convenience.

Club Motion. US8951138B2. This disclosure addresses club path and face orientation around impact. Its inclusion supports discussion of club measurement methods, not a blanket assertion that four cameras or reflective face dots are universally protected, or that one patent identifies a particular commercial club-data system.

7.3 FlightScope / EDH — Henri Johnson

Phase-Modulated Spin Signals. US9868044B2. Claim 1 derives spin from the period of a demodulated radar component whose portions correspond to apparent speeds above and below a nominal speed [27]. The disclosure’s dielectric-lens explanation and surface-feature examples should be distinguished from that claim’s signal-processing requirements. Symmetry can cause a feature pattern to repeat more than once per revolution; a measured modulation frequency therefore needs interpretation before it becomes a spin rate.

Receiver-Pair Timing. US10775492B2. Claim 1 uses demodulated Doppler signals from receiver pairs in perpendicular planes, their relative time delays and receiver separations to calculate an axis relative to a receiver plane [26]. This supplies rotation-related observations without first fitting a long curved trajectory. It does not establish unconditional indoor accuracy: signal quality, delay estimation, geometry and the stated axis convention remain part of the result. A ratio of delays also needs careful treatment near a vanishing denominator and when resolving angle quadrants; a bare one-argument arctangent is not a complete implementation.

Camera/Radar Tracking. US10338209B2. The document describes calibrated simultaneous tracking and combining camera and radar information [25]. Claim 1 specifically includes using radar data to interpolate a three-dimensional camera trajectory when one sensor fails to supply usable measurements. Recovery through such a gap is an inference step whose uncertainty must be tested, not a second independent observation of the missing interval.

7.4 Acushnet (Titleist) — Optical Measurement Architectures

These documents illustrate several ways to assemble an optical launch monitor [17, 16, 30]. They do not reduce to one fixed camera-and-marker recipe.

Marked Club and Ball. US5501463A. Claim 1 uses at least two cameras and repeated exposures of contrasting areas on both the striking instrument and the object. Image patterns support estimation of club path and orientation. The conversion from marker positions to a club-face quantity requires a defined relationship between those markers and the club geometry.

Launch Conditions to Trajectory. US6500073B1. Claim 1 obtains positions and orientations from images at different times, calculates launch conditions, then calculates a trajectory. Measured initial conditions and the resulting modeled flight must be distinguished: a predicted landing point is not an observed landing point.

Separate Club and Ball Capture. US6758759B2. The first claim places club and ball cameras in vertically separated planes and derives their motion data from repeated images. The minimum in that claim is one camera for each task; describing the entire patent as requiring two stereo pairs would confuse an embodiment with the claim.

Integrated Optical Hardware. US7143639B2. Claim 1 specifies a housing layout with a club camera, at least two ball cameras, illumination, triggering and a teeing aid, arranged for use by both handednesses through rotation of the apparatus. The first claim does not require four cameras.

Three-Dimensional Image Generation. US10668350B2. The disclosure discusses stereo and light-field approaches. Claim 1 specifically includes a light-field camera that captures light intensity and ray direction, followed by reconstruction and kinematic processing. Example frame rates or exposures in a disclosure are design proposals; they do not demonstrate a product’s achieved measurement error.

7.5 United States Golf Association — Aerodynamic Experiments

The United States Golf Association’s US6186002B1 belongs outside the Acushnet grouping. It describes controlled launches and timed light-screen crossings for aerodynamic coefficient fitting (Chapter D) [34]. Its experimental setup matters: an arbitrary short trajectory need not contain enough information to separate initial spin, wind and uncertain aerodynamic coefficients.

7.6 Korean Ecosystem: Creatz/Uneekor and Golfzon

Uneekor’s marking notice connects specific products with listed patent numbers [76]. The Creatz disclosures include US10247553B2, which describes a start sensor for identifying ball placement and starting movement, and US12008770B2, which concerns dimple-based spin estimation. A product’s markerless-spin label does not specify its entire algorithm or establish performance across every ball surface.

Creatz’s US11191998B2 makes the distinction between observation and inference particularly clear. Claim 1 uses speed- or angle-related quantities to estimate spin when image marks cannot supply it; when mark-based calculation is possible, it also uses such an estimate to correct that calculation. This is more than a fallback for missing observations. The reported spin can depend on both image evidence and a relationship inferred from other ball or club quantities. Validation should distinguish these operating modes and test their errors separately; a plausible corrected value does not establish an independent measurement of rotation.

Golfzon’s US12002222B2 provides a different example of this dependence. Claim 1 obtains dynamic loft and backspin through a reference database using observed club/ball motion parameters, then calculates spin-axis and side-spin quantities from motion vectors and specified relationships. These inferred outputs must not be used as independent evidence that the assumed impact relationships hold for a particular shot.

Golfzon’s US9242158B2 describes beginning the displayed flight with a first set of ball information and updating it when a second set arrives [81]. This separates feedback latency from completion of the measurement pipeline. Timing examples in a description are not measured latency guarantees. Readers comparing monitors should ask both when an output appears and when the final parameters become available.

7.7 Technical Research Map and Rights Boundaries

The following map identifies engineering questions to investigate. It replaces a technique-wide “free to use” classification: the age or expiration of one document cannot establish clearance for every implementation bearing the same technique name. The USPTO describes a patent as a right to exclude, not permission to practice an invention. Term cannot be computed indiscriminately as twenty years from any date labeled “priority”; the applicable filing basis and other term conditions matter.¹ A design-specific rights review requires the relevant claims, jurisdictions and current records. This technical catalog does not supply that determination.

Table 7.1: Selected Measurement Methods and Their Validation Questions

Method	Example Documents	Question to Test
Surface-image spin	US7292711B2; US7324663B2	Are texture, exposure and rotation matching sufficient to resolve ambiguity?
Marked club/ball images	US5501463A; US6758759B2	Are calibration and marker-to-face geometry known at the reported time?
Radar spin periodicity	US8845442B2; US9868044B2	Which spectral or temporal repeat corresponds to one revolution?
Radar axis information	US10850179B2; US10775492B2	Is the result a projection or a full axis, and in which coordinate frame?
Camera/radar fusion	US10338209B2; US10989791B2	Are clocks, coordinates, association and missing-data uncertainty controlled?
Trajectory-informed spin	US12544624B2	How strongly does the result depend on club probabilities and the flight model?
Placement and triggering	US7641565B2; US10247553B2	Does detection reliably capture the interval needed for the requested output?
Staged display	US9242158B2	Which values are provisional, and when does the displayed trajectory change?

Key point

A launch monitor is a chain of observations and inferences. Radial speed, image displacement, surface rotation, launch conditions and predicted flight occupy different places in that chain. Combining sensors can add information, but only after timing, coordinates and object identity agree; it does not turn repeated use of the same data into independent evidence.

For example, Topgolf’s US12544624B2 uses trajectory observations, spin candidates and probabilities associated with a likely club. That approach differs from tracking rotating surface features. A plausible spin value may partly reflect the prior model, so evaluating both methods requires experiments that expose their different ambiguities.

When using these outputs to investigate a swing, retain the measurement frame, time interval, calibration and model assumptions. Launch conditions constrain the ball’s initial state; they alone do not identify the golfer’s joint torques, force history or unique movement strategy. Technical rigor means carrying those information limits through the argument, rather than letting a precise display imply a uniquely determined cause.

¹USPTO, *Patent Essentials*, <https://www.uspto.gov/patents/basics/essentials>.

Chapter 8

Ball Flight Models and Trajectory Estimation

Both sensing families depend on an aerodynamic model: cameras integrate it forward from measured launch to get carry; radars fit it to partial trajectories to extract spin axis and to extrapolate indoor flights.

8.1 Equations of Motion

With air density ρ , ball radius $R = 21.34\text{ mm}$, mass $m = 45.93\text{ g}$, cross-section $A = \pi R^2$, and airspeed $\mathbf{v}_a = \mathbf{v} - \mathbf{v}_{\text{wind}}$:

$$m \frac{d\mathbf{v}}{dt} = -\frac{1}{2}\rho AC_D \|\mathbf{v}_a\| \mathbf{v}_a + \frac{1}{2}\rho AC_L \|\mathbf{v}_a\|^2 (\hat{\boldsymbol{\omega}} \times \hat{\mathbf{v}}_a) + m\mathbf{g}, \quad (8.1)$$

with the drag and lift coefficients functions of Reynolds number $\text{Re} = 2R\|\mathbf{v}_a\|/\nu$ and spin ratio $S = R\omega/\|\mathbf{v}_a\|$. Fourth-order Runge–Kutta integration is ample; carry is evaluated where the trajectory returns to launch elevation. Tilting the spin axis by θ rotates the Magnus force off vertical, giving lateral acceleration $(F_M/m)\sin\theta$ — the mechanism behind the 0.7%-per-degree side-curve rule and Equation (2.2).

8.2 Aerodynamic Coefficient Data

Bearman & Harvey (1976) [2]. The canonical wind-tunnel dataset on spinning golf-ball models across the flight envelope. Dimples drop the critical Reynolds number to $\sim 5 \times 10^4$; post-critical $C_D \approx 0.25$ is nearly Re-independent; C_L rises with spin ratio from ≈ 0.08 to 0.25 over $S \approx 0.02$ –0.3; hexagonal dimples outperform round ones.

Smits & Smith (1994) [54]. The parameterized model most widely used in simulators:

$$C_D = C_{D1} + C_{D2} S + C_{D3} \sin(\pi \frac{\text{Re}-A_1}{A_2}), \quad C_L = C_{L1} S^{0.4}, \quad (8.2)$$

with $C_{D1} = 0.24$, $C_{D2} = 0.18$, $C_{D3} = 0.06$, $A_1 = 9 \times 10^4$, $A_2 = 2 \times 10^5$, $C_{L1} \approx 0.54$, plus a spin-decay law with time constant $\approx 24\text{ s}$ at 100 mph (roughly 4%/s early in flight).

Quintavalla / USGA Indoor Test Range (2002) [49]. Six-term polynomial C_D/C_L models in Re and S fitted from trajectory photography on the USGA ITR; captures the low-speed end-of-flight regime better than wind tunnels; the basis of USGA Overall Distance Standard conformance testing. The companion method patent US6186002 [34] — determining coefficients from measured trajectories — is the template for calibrating an open model against real flights.

Design implication

An implementation should ship Smits–Smith (Equation (8.2)) with spin decay as the default flight model, structured so a Quintavalla-style six-term fit can drop in, and calibrate against measured outdoor trajectories (or MLM2PRO reference data) using the US6186002 trajectory-fitting approach. Carry disagreements between simulators are dominated by model choice, not launch measurement — version the model and record it with every shot log.

8.3 Trajectory Estimation and Filtering

The estimation core of a radar launch monitor is a nonlinear filtering problem. State $\mathbf{x} = (\mathbf{p}, \mathbf{v}, S, \theta_{\text{axis}})$; process model = Equation (8.1); measurement models: radar $h(\mathbf{x}) = (r, \text{az}, \text{el}, \dot{r})$ with $\dot{r} = \mathbf{v} \cdot \hat{\mathbf{p}}$, camera $h(\mathbf{x}) = \text{pixel projections}$. An extended (or unscented) Kalman filter runs the standard recursion; launch parameters are then obtained by *smoothing and back-extrapolation* — fit the entire observed arc and evaluate the state at the moment of face separation, which is far more robust than differencing the first noisy fixes. Including θ_{axis} in the state makes spin-axis-from-curvature (Section 4.4.3) fall out of the filter naturally, and per-measurement confidence weighting is the clean way to fuse heterogeneous sensors (OPS243-A speed, K-LD7 angles, future camera fixes).

Design implication

A two-module radar design correlates angle-radar bursts with the CW impact timestamp and reads angles from single detections. Migrating to a short EKF smoother over the full K-LD7 ring-buffer burst — even 10–20 detections over 50 ms — would use all available data, reject outlier bins, and yield launch angles with quantified covariance. The same filter skeleton later absorbs camera measurements unchanged.

8.4 Indoor Extrapolation and Its Errors

Indoors, every system integrates Equation (8.1) from launch conditions. Error propagation is dominated by spin uncertainty: at driver speeds, ± 300 rpm maps to roughly ± 4 – 6 yd of carry and ± 1 yd of side; a $\pm 2^\circ$ spin-axis error at 2,500 rpm maps to ± 3 – 4 yd of side at 250 yd carry. This is why the accuracy literature (Chapter 9) finds spin to be the fragile channel indoors for radar, and why camera systems — which measure spin directly — win the indoor comparison structurally.

Chapter 9

Accuracy: The Independent Evidence

9.1 Peer-Reviewed Validation

Leach, Forrester, Mears & Roberts (2017) [31] remains the key criterion study: 240 shots (driver, 7-iron, wedge) measured simultaneously by a TrackMan Pro IIIe, a Foresight GC2+HMT, and a four-camera 5,400 fps optical reference. Findings: ball parameters agreed well on both devices (TrackMan clubhead-speed median difference -0.4 mph; ball speed $+0.2$ mph; launch angle 0.0°), but **club parameters were materially worse for both** — the authors endorse ball data for research use and advise caution on club data. This is the empirical face of the directness hierarchy of Section 2.3: what is measured agrees; what is derived diverges.

TrackMan 4 indoor reliability (J. Sports Sciences, 2024) [82]: within- and between-session reliability in high-level golfers indoors — ICC 0.99 for club speed, 0.97–0.99 for ball speed, but **spin-rate ICC as low as 0.02–0.60**: even the reference radar’s indoor spin channel is fragile when flight is truncated.

A 2025 Mevo+ vs. TrackMan 4 indoor agreement study exists [1] (abstract-level access only at review time), extending the same pattern down-market.

9.2 Robot and Industry Testing

Golf Laboratories robot testing (reported via Foresight and third parties) puts GCQuad center-strike spin standard deviation at ≈ 82 rpm versus ≈ 175 rpm for TrackMan 4 — the photometric spin advantage in its clearest form [20]. MyGolfSpy’s indoor comparison against a GCQuad reference found the camera-assisted MLM2PRO among the tightest budget units for launch, spin, and carry, with radar-only units (R10, original Mevo) trailing indoors [39]. Manufacturer-quoted specs bracket the market: ± 1 mph ball speed and $\pm 1^\circ$ launch angles are common claims; club-face claims ($\pm 2^\circ$, “calculated”) are visibly weaker.

9.3 What Drives Inter-Device Disagreement

Synthesizing the validation literature and the architecture analysis:

1. **Reference-point differences** in club speed (geometric center vs. fastest return) — several mph of systematic spread.
2. **Derived face data**: D-plane inversion amplifies launch-direction bias by $\sim 1.15\times$ and cannot see gear effect, so radar face angle diverges from optical face angle most on off-center strikes.
3. **Indoor spin**: sideband SNR collapses on short flights and clean balls; systems silently switch to estimation, and estimated spin feeds the carry model.

4. **Flight-model differences:** identical launch conditions produce different simulated carries across vendors; this is a modeling disagreement, not a measurement one.
5. **Alignment:** a unit misaligned to the target line biases path, face, and launch direction coherently — the cheapest error to fix and the most common in practice.

Design implication

For validation against a consumer unit such as the MLM2PRO (a common reference instrument): compare ball speed and launch angles directly; compare spin only on RPT/RCT marked balls where the MLM2PRO's measurement is camera-based; expect club-speed offsets from reference-point differences and calibrate a per-club correction rather than chasing agreement; and log raw I/Q captures so spin-detection improvements can be replayed against historical shots.

Chapter 10

Design Guidance for Implementers

This chapter distills the review into design guidance, organised by what a given sensing architecture can and cannot deliver. It is written for anyone building a launch monitor, and equally for anyone evaluating one: the same reasoning that tells an engineer which parameters are reachable tells a buyer which reported numbers deserve trust.

The baseline throughout is a **minimal radar architecture** — a single continuous-wave Doppler module for speed, one or two interferometric angle-radar modules for direction, a hardware impact trigger, and an embedded host. Representative parts are named where they are useful (the OmniPreSense OPS243-A at 24 GHz, the RFbeam K-LD7, a Raspberry Pi class host), but nothing here depends on those specific choices. This configuration is a minimal but honest instance of the commercial radar architecture, and it is the cheapest starting point that measures anything directly.

10.1 Ball Data

1. **Ball speed** is the strongest channel in any radar design: the Doppler line is unambiguous, and $\pm 0.5\%$ class accuracy from a commodity module matches commercial practice. Apply the aspect-angle (cosine) correction using the measured launch angles — at 3–5 ft behind the tee with a driver launch of 12° the radial underestimate is small but *systematic*, and systematic errors are the ones that survive averaging.
2. **Launch angles**: treat angle-radar bursts as a short trajectory, not a single detection. An EKF smoother over the ring buffer (Section 8.3) with back-extrapolation to the impact timestamp is the single highest-leverage software upgrade for angle quality, and it costs nothing in hardware.
3. **Spin rate** is where radar earns its reputation for inconsistency. A $\sim 50\text{--}60\%$ detection rate from I/Q sideband analysis is consistent with the physics rather than evidence of a bug: sideband SNR depends on ball-surface asymmetry and observation time (Section 4.4.2). Improvements in order of cost: longer observation windows (later trigger cutoff); cepstrum or harmonic-product comb estimation with equal-spacing qualification; marked-ball guidance for indoor use; and honest fallback — report estimated spin from club type, speed and loft priors, clearly flagged, which is what Garmin does and documents. *Disclosure reference*: US8845442 describes radar spin processing; the technical map in Section 7.7 does not determine clearance or a technique-wide expiry date.
4. **Spin axis**: short flight segments may provide too little trajectory information to separate axis, aerodynamic coefficients and launch-condition errors reliably. Rotation-related observations from receiver geometry provide another disclosed route (Section 7.3), but require suitable hardware and validated signal processing. Optical rotation estimates and explicitly labeled impact-model estimates are further options; their observation requirements and error sources differ.

10.2 Club Data

1. **Club speed:** define and document the reference point — this is the single most consequential and most neglected decision in the whole design (Section 2.1.1). A CW module sees a smear of head and shaft returns; gating the pre-impact spectrum and taking a fixed percentile of the velocity distribution, rather than the maximum, approximates a stable reference and avoids the toe-speed inflation that plagues naive processors (Section 4.5). Validate smash factor against the loft-appropriate ceiling (Section 3.2) and flag violations rather than displaying them.
2. **Club path** and **attack angle** from angle radar are legitimate direct measurements — the same primitives a consumer radar unit measures. Alignment calibration dominates their accuracy, and no amount of signal processing recovers a misaimed sensor.
3. **Face angle:** the D-plane inversion (Equation (4.6)) with the loft-dependent weight (Equation (3.3)) is exactly what radar-only commercial units report. Implement it — but label it *derived*, propagate its error (a 1.32× amplification of launch-direction bias at the measured weight; see the caution in Section 3.1.1), and suppress it on detected off-centre strikes once impact location is available. The D-plane forward model doubles as a simulator’s launch generator, so one well-tested module serves both directions.
4. **Impact location:** club and ball imaging offers a concrete route to locating contact relative to the face. The Acushnet and Wintriss disclosures (Sections 7.2 and 7.4) illustrate measurement architectures, rather than complete validated implementations or blanket rights clearance. Face fiducials can constrain orientation through calibrated image geometry and a known marker-to-face relationship. Camera count alone does not establish three-dimensional observability, visibility at impact or achieved accuracy.

10.3 Capability Tiers

The following tiers are cumulative. Each is a coherent product in its own right, and each is defined by what it moves from derived to measured rather than by a parts list.

1. **Tier 1 — radar only.** EKF smoothing, comb-based spin with honest fallback, D-plane-derived face data with provenance flags, smash-factor sanity gates, and a documented alignment-calibration procedure. This delivers the consumer-radar feature set. Everything club-face-related is inferred.
2. **Tier 2 — add optical spin, then impact imaging.** One or two global-shutter cameras with an IR strobe, at modest incremental cost. Calibrated image sequences can support launch-angle and surface-rotation estimation; impact-location capture additionally needs a suitable view of the club and ball. Radar may contribute triggering, speed and trajectory observations, depending on the sensor arrangement. The disclosures in Chapter 7 supply design comparisons, not a guarantee that this combination is accurate or cleared for a particular implementation.
3. **Tier 3 — true fusion.** A single EKF consuming radar speed, radar angles and camera fixes with per-sensor covariances; a flight model per Chapter 8 with versioned coefficients; and a validation protocol per Chapter 9. The versioning matters more than it sounds: Chapter 8

shows that a coefficient difference of 0.01 in C_D is worth roughly eight yards, so an unversioned model change is a silent recalibration of every number the device has ever reported.

4. **Tier 4 — measured club delivery.** An imaging radar with custom chirp firmware feeding the screw-theoretic rigid-body estimator of Chapter E: per-detection Doppler rows solved for the club’s twist, smoothed on $SE(3)$, and projected into club speed at a declared reference point, measured path and attack angle, closure rate, and an ISA-defined swing plane. This moves path and attack angle from *derived* to *measured* in Table 2.3. Face angle and impact location remain assigned to the optical module in this proposed architecture; the allocation is a design choice, not proof that every other sensing approach is impossible.

10.4 Reporting Honesty as a Design Feature

The clearest lesson of this review is not about sensors at all.

Commercial marketing blurs the measured/derived boundary, and the independent validation literature repeatedly punishes the derived quantities: clubhead velocity met research-grade tolerance on 54% and 29% of shots for the two devices ever tested against a traceable optical benchmark, and club orientation data was returned on only 62% of shots overall — 19% for a utility wedge (Chapter 9). None of that appears on a specification sheet.

An instrument can therefore differentiate itself by doing the opposite: tagging every reported parameter with its provenance (measured, derived, or estimated), its uncertainty, and the model version used, and reporting its own per-shot success rate rather than only its best-case tolerance. Table 2.3 is effectively the schema for that tagging.

Two vendors already demonstrate that this is commercially survivable rather than suicidal. Garmin publishes an explicit measured-versus-calculated split with tolerances on each side, and states outright that its face angle is algorithmic. TrackMan publishes the reference-point discrepancy between its own quantities and warns that numbers from different methodologies “aren’t comparable.” Both remain market leaders in their segments. The honest disclosure did not cost them anything, and it is the closest thing this industry has to a standard worth adopting.

Chapter 11

Governed Analytics and Validation Program

The distinction between a useful analytics platform and a scientific vendor comparison is the evidence design. A large collection of launch records can support exploratory model checking without supporting the stronger claim that one device disagrees with another. The repositories associated with this review therefore use a two-release program with a versioned data contract, explicit use eligibility, and fail-closed statistics. The program is tracked by the public Tools program epic. UpstreamDrift is the canonical generic contract and evidence-envelope authority. Tools owns the current specialized grouped estimators: Python is the desktop calculation authority and React is a contract-tested browser twin. Those grouped operations are not yet UpstreamDrift API operations, so that boundary is published rather than obscured by a cross-service parity claim.

11.1 Qualified Corpus Snapshot

The controlled authority snapshot contains 261,666 rows from 27 traceable sources. Size alone is misleading. Only 13,855 rows contain finite, non-imputed values for all five primary launch inputs (ball speed, launch angle, launch direction, spin rate, and spin axis). Before the per-source minimum-size gate, 13,647 rows also contain carry, 13,532 lateral outcome, 9,850 descent angle, and 9,283 flight time. The strict model cohort contains 13,843 rows from 15 source families because each predicted outcome has its own availability rule. Eligibility is evaluated by source, metric, provenance, and intended use; a blank field is represented as *unavailable*, never silently filled merely to increase sample size.

The largest source is a 236,590-row ShotLink-derived table. It has no complete five-input launch rows and no observed carry values. Its rights and transformation lineage also require separate approval. It is therefore quarantined: the source is **not eligible for vendor-model training**, public redistribution, or vendor-performance claims. It may be retained internally for provenance review and for analyses that are independently shown to match its permitted use. This rule prevents a large but unsuitable table from dominating the apparent evidence base.

At the present qualification gate, no vendor-labelled surrogate is currently eligible for production training. The corpus contains 11,699 TrackMan-labelled rows and 9,298 strict launch-input rows, but it lacks a trusted repeating player or session grouping. Foresight has four rows (two strict); FlightScope has 2,794 rows but no strict five-input rows. The historical TrackMan surrogate used unique shot identifiers as partitions, which cannot protect against player, session, or source leakage; it is retained for audit with status **retired_non_group_safe** and cannot be loaded, queried, or retrained. A future model that passes the group-safe gate would be described as *vendor-comparable on its qualified source*, not as a reproduction or certification of proprietary vendor software.

Key point

Every displayed statistic carries its source identity, row-selection rule, units, method version, and backing row references. An unavailable result is a valid result: it records why the requested inference is not supported.

11.2 Release A: Qualified Analytics Platform

Release A delivers the professional analysis surface on data that passes the declared gate. Its scope includes unit-aware scatter and distribution views, model residuals, arbitrary variable relationships, within-player and between-player covariance, session trends, uncertainty, exportable plots, and reproducible project files. Player-level analysis requires an explicit player identity column. Identity is never inferred from session, club, row order, filename, or directory layout; missing and whitespace-only identities are excluded and reported.

UpstreamDrift’s versioned contract returns estimates, uncertainty, warnings, source lineage, backing-row references, and machine-readable unavailable states for its registered generic analyses. Tools supplies PyQt and React/Vite views with parity in variable choices, units, formulas, backing-data export, persistence, and explanatory text. The specialized grouped estimators for covariation and longitudinal synthesis are currently released within Tools, with Python as the desktop authority and TypeScript as a tested browser twin; they are not yet UpstreamDrift API operations. This bounded duplication is a tracked architectural seam, not evidence that two independent implementations are automatically equivalent.

Exploratory all-pairs scans report multiplicity warnings. Within-player analysis separates raw pooled association, player-mean-centred association, between-player association, and per-player estimates; these quantities answer different questions and must not be substituted for one another. Correlation and principal-component loadings are descriptive, not causal. Predictive work uses grouped splits whenever a trusted player or session identity exists and otherwise states that leakage-resistant grouping is unavailable.

Strokes gained is similarly gated. A result labelled *strokes gained* requires a cited baseline table, a declared lie and distance state, and a reproducible mapping from shot outcome to expected strokes. A distance-only or target-error score without that backing is labelled a *performance proxy*; it must not borrow the strokes-gained name.

The canonical scoring contract requires exact lie, context, target, and distance states before and after the shot. Missing states, incomplete retained rows, and queries outside baseline support fail closed. No expected-strokes baseline is bundled with the public application. A baseline’s declared source URL, licence, version, and digest preserve provenance; that declaration does not independently endorse its methodology, legal suitability, or relevance to a particular player population. The distance/target performance proxy remains a separate yard-based dispersion measure whose machine-readable flag is `is_strokes_gained=false`.

11.3 Qualified Model-Comparison Results

The Release A campaign evaluated seven open flight-model configurations on the 13,843-row strict cohort, producing 96,901 predictions with no execution failures. Its primary estimands are source-stratified error, bias, association, and residual relationships. Pooled metrics are retained only as descriptive summaries because source composition, device, venue, ball, and acquisition conditions are confounded.

For the Blackmore TrackMan-labelled carry cohort ($n = 8,860$), the best tested configuration was the Waterloo/Penner implementation: carry RMSE 12.256 m, mean error -4.093 m, and strong linear association with the reported outcome. These are agreement statistics against the source's reported carry, not an absolute-accuracy result and not evidence that the proprietary TrackMan equations have been identified. For context, the best tested carry configuration on one Garmin R10 source ($n = 1,541$) had RMSE 7.150 m and bias -4.722 m, while one Garmin R50 source ($n = 255$) had RMSE 10.238 m and bias $+6.554$ m. The sign change is one reason a pooled vendor ranking would be scientifically unsafe.

The content-addressed calculation record is the private campaign's `results/v2/model/model_campaign_manifest.json`. It binds the cohort, predictions, source metrics, pooled descriptive metrics, residual-correlation table, configuration, and source qualification outputs by SHA-256. The companion `multisource_v2_model_report.md` states row selection and limitations in human-readable form. Restricted rows remain in the private authority; the public paper reports only qualified aggregates.

The qualified snapshot is pinned to private authority commit and three independently verifiable digests:

```
private merge: 8cd8ad04bf2904eedd0d8f10d2aa202e437e91fd
qualification: 679856edb290111a0106c0bcbd2b673f581fd6b0026409c4d643565b330709a
model campaign: b70935fb4209b3f3aa18c3129ae948c4e20da74536af4d88433e5f0875960ca7
capability manifest digest: 1906d19fcace3284ae99d9dd8de213a0e2dabe8c062e4d63edc06c98f7eaf92e
```

These identifiers bind the reported aggregates without exposing restricted shot rows.

The associated software and public-boundary revisions are pinned separately:

```
public data facade: 70feeb4889ef4d030bdb426bd7ecb465c36ba41b
generic UpstreamDrift contract v2: 5ea0d08821ee8a1cacb93d22239abddee4813fca
Tools grouped-analytics release: 08b4b013d8f58feb0a788fc67add29f64750da23
canonical source-backed SG contract: fb4e6e41ac4980309e32dacceb957f4da02616b3
Tools canonical SG integration: a7b6dc437b45857e27cd07cccb56ff665826e342
Tools visual approval: 10d7f6fc38d27dd04324160090a7290f46d0cfd4
approved launch-monitor PNG SHA-256:
6699129037d0adf8152e73bdae3fc2b75a2e15bfb16e25a4b3cfcec2448ac00e
```

The public facade contains capability metadata and locks, not restricted rows. The older Tools revision remains the authority for the specialized grouped estimators, while the older UpstreamDrift revision remains the generic v2 envelope and claim-discipline authority. The newer UpstreamDrift revision owns the canonical source-backed scoring contract, and the newer Tools revision is its PyQt/React consumer. The visual approval covers presentation and accessibility, not statistical correctness or baseline validity. It also does not convert an unbundled or user-selected expected-strokes table into endorsed scientific evidence.

11.4 Release B: Paired-Device Scientific Validation

Release B begins only after a preregistered paired-device protocol has been executed. The same physical shot must be observed simultaneously by each device under comparison, with device model and firmware, ball and club, indoor/outdoor condition, environmental state, alignment, player identity, shot identity, and measured-versus-estimated status recorded. Randomized blocks should span speed, loft, spin, launch direction, impact location, and representative mishits. A traceable reference system is required where a criterion claim is made.

The primary estimands are paired bias, limits of agreement, error as a function of magnitude and condition, repeatability, missingness, and device-by-condition interaction. Bland–Altman analysis and hierarchical models are more informative here than correlation alone: two devices can be highly correlated while disagreeing systematically. Sample-size targets and exclusion rules must be fixed before inspecting device differences. Firmware changes form new strata rather than being pooled invisibly.

The preregistered confirmatory design uses exactly three robot-centre, outdoor club/speed cells with 84 accepted shot pairs per cell (252 total). Each pair requires observations from two devices and exactly one independent reference; calibration must pass, device rows must be unique within a pair, serial identifiers must be SHA-256 hashes, and all members share the common trigger timestamp. The protocol, capture-row JSON Schema, Python validator, deterministic 252-pair schedule, immutable collection ledger, command-line procedure, and synthetic fixture are complete. The generated status record reports `confirmatory_ready=false`, 252 `not_collected` exclusions, and zero analyzed pairs. Its schedule and ledger are independently bound by SHA-256:

```
schedule: 72b06c034e73751dbacce8468505517a8500754d4d769cc89fed846611790173
ledger:   ebc1a24100b0e09ea39978127d6f1bcad3bd26bf6c4f3ae042680a833281d9c9
```

Because the paired observations have not yet been collected, protocol readiness is not reported as a completed device-validation result; the status is a software hold point, not device evidence.

Neural-network vendor emulators remain internal research artifacts. Their training manifests must pin the eligible sources, feature and target units, splits, random seeds, software version, and evaluation cohort. They are released only with held-out and out-of-distribution performance, uncertainty or applicability bounds, and a model card that forbids device-emulation or certification claims. Until the paired-device gate is satisfied, Release B comparisons remain unavailable rather than being approximated from unrelated shots.

11.5 Reproducibility and Claim Discipline

The repository fleet records exact Git revisions for the statistical authority, user interfaces, paper, public data facade, and private data authority. A release manifest binds those revisions to the data manifest and its checksums. Source files remain immutable; normalization and exclusions are transformations with their own lineage. Project exports contain both the request and complete backing rows so another analyst can reproduce the displayed result without relying on GUI state.

These controls deliberately narrow the claims. Public launch-monitor records can compare an open flight model with reported outcomes under stated source conditions. They cannot, by themselves, identify a proprietary vendor model, separate sensor error from aerodynamic-model error, or establish cross-device agreement. Those questions belong to Release B and require paired evidence.

Appendix A

Annotated Reference Library

This appendix organizes sources used in the review by their evidential role. The research dossiers in `articles/Launch_Monitor_Technology_Review/research/` record the working claim trails; they are research notes, not independent validation. A source listed here does not support every claim about the associated product. Manufacturer descriptions, measured agreement, and physical models answer different questions. Read a numerical result together with its device version, measurand, reference point, event time, test conditions, and missing-data rate. Access checks on 1 October 2026 are observations, not availability guarantees: a failed retrieval does not establish permanent absence.

A.1 Patents (Google Patents / USPTO)

These records disclose proposed methods and embodiments. They do not establish the architecture shipped in a particular device, independent performance, or freedom to implement. The annotations identify technical topics; they do not determine claim scope or current legal status. Inventor and company groupings are navigation aids, not a complete ownership or portfolio history.

TrackMan A/S (Fredrik Tuxen)

- Inventor index and manufacturer patent list:
<https://patents.justia.com/inventor/fredrik-tuxen> and
<https://www.trackman.com/legal/patents>
- US8845442 — Determination of spin parameters of a sports ball (harmonic-sideband spin; Magnus spin-axis inversion): <https://patents.google.com/patent/US8845442B2/en>
- US10393870 — continuation of the above:
<https://patents.google.com/patent/US10393870B2/en>
- US10850179 — spin axis via receiver-pair interferometry:
<https://patents.google.com/patent/US10850179B2/en>
- US8085188 / US9857459 / US10473778 — target-deviation (tap-a-target) family:
<https://patents.google.com/patent/US8085188B2/en>,
<https://patents.google.com/patent/US9857459B2/en>,
<https://patents.google.com/patent/US10473778B2/en>
- US10315093 — trajectory tracer overlay:
<https://patents.google.com/patent/US10315093B2/en>
- US11086005 — multi-bay range tracking:
<https://patents.google.com/patent/US11086005B2/en>
- Historical TrackMan spin-patent explainer address (redirected to the general blog during this review; use the patent records above for content):
<https://blog.trackmangolf.com/patent-measuring-the-spin-of-a-sports-ball/>

FlightScope / EDH (Henri Johnson)

- US9868044 — spin from periodic features in a phase-demodulated signal:
<https://patents.google.com/patent/US9868044B2/en>
- US10775492 — spin-axis estimation using receiver-pair time delays:
<https://patents.google.com/patent/US10775492B2/en>
- US10338209 — combined radar and camera tracking:
<https://patents.google.com/patent/US10338209B2/en>
- Company history: <https://athlonsports.com/golf/the-flightscope-story-tracking-the-history-of-flight-in-golf>

Foresight Sports / Wintriss Engineering (Kiraly)

- US7292711 — flight parameters from imaged surface features, including markings, blemishes, or dimples: <https://patents.google.com/patent/US7292711B2/en>
- US7324663 — standalone smart-camera launch monitor:
<https://patents.google.com/patent/US7324663B2/en>
- US7497780 — integrated launch monitor UX:
<https://patents.google.com/patent/US7497780B2/en>
- US7641565 — ball-placement detection: <https://patents.google.com/patent/US7641565B2/en>
- WAWGD (Foresight) filings index:
<https://patents.justia.com/inventor/wawgd-inc-dba-foresight-sports>
- Foresight–Uneekor 2024 license (Business Wire):
<https://www.businesswire.com/news/home/20240930838753/en/>

Acushnet / Titleist (Gobush et al.)

- US5501463 — optical club/ball measurement with reflective markers:
<https://patents.google.com/patent/US5501463A/en>
- US6500073 — camera measurements and trajectory computation:
<https://patents.google.com/patent/US6500073B1/en>
- US6758759 — separate club and ball monitors before/after impact:
<https://patents.google.com/patent/US6758759B2/en>
- US7143639 — portable monitor with one or more cameras:
<https://patents.google.com/patent/US7143639B2/en>
- US8500568 / US8556267 — portable continuations:
<https://patents.google.com/patent/US8500568B2/en>,
<https://patents.google.com/patent/US8556267B2/en>
- US10668350 — stereographic or light-field camera embodiments:
<https://patents.google.com/patent/US10668350B2/en>

Others

- USGA US6186002 — aerodynamic characteristics from measured trajectories (original assignee: United States Golf Association): <https://patents.google.com/patent/US6186002B1/en>
- Full Swing US2020/0147470 — CW+FMCW dual-mode launch monitor: <https://patents.google.com/patent/US20200147470A1/en>
- Creatz US10247553 — start-position sensor: <https://patents.google.com/patent/US10247553B2/en>
- Golfzon US9242158 — two-stage latency-hiding simulation: <https://patents.google.com/patent/US9242158B2/en>
- Uneekor patent/license list: <https://uneekor.com/legal/patents>
- Pulse/exposure timing correlation US12401909: <https://patents.google.com/patent/US12401909B2/en>
- TrackMan EP1698380 litigation coverage: <https://thegolfwire.com/322771-2/>
- FlightScope 2022 BGH win vs TrackMan: <https://golfbusinessnews.com/news/innovation-centre/flightscope-wins-patent-infringement-case-against-trackman-in-germany/>

A.2 Regulatory Records and Teardown Reports

FCC filing mirrors identify equipment records and tested configurations; they do not independently validate launch parameters. Forum teardown reports are secondary observations requiring confirmation against the actual hardware.

- TrackMan 4 FCC ID SFX-TMAN4 (X-band + 24 GHz): <https://fccid.io/SFX-TMAN4>
- TrackMan iO FCC ID SFX-TMB0201 (24 GHz): <https://fccid.io/SFX-TMB0201>
- FlightScope Mevo FCC ID QXP-A7310 (24.125 GHz): <https://fccid.io/QXP-A7310>
- Mevo+ teardown / QXP-LJ361 (10.5–10.55 GHz) discussion: <https://golfsimulatorforum.com/forum/flightscope/277362-mevo-teardown>

A.3 Manufacturer Documentation and Product Reporting

Official pages describe vendor definitions and specifications. Media, retailer, and community pages below are labeled separately; their inclusion does not turn a product claim into an independent measurement.

TrackMan

- 40+ parameters explained: <https://www.trackman.com/blog/golf/40-trackman-parameters>
- Club data definitions: <https://www.trackman.com/blog/golf/club-data-definitions>
- Club speed (geometric-center definition): <https://www.trackman.com/blog/golf/what-is-club-speed>
- Spin loft / spin rate: <https://www.trackman.com/blog/golf/spin-loft>, <https://www.trackman.com/blog/golf/spin-rate>

- OERT (“Two radars, one camera, zero doubt”):
<https://www.trackman.com/blog/golf/two-radars-one-camera-zero-doubt>
- Tech specs (TM4, iO): <https://www.trackman.com/golf/launch-monitors/tech-specs>
- Ball Flight Laws newsletter (Jan. 2009; historical D-plane and launch-direction teaching material). Compare the study-specific weights in §3.1.1; 0.76 is not a universal constant:
<https://www.yumpu.com/en/document/view/11652756/trackman-ball-flight-laws>

A.3.1 The TrackMan Newsletter Archive

Historical newsletters are indexed below through a third-party mirror. Issues #7–#9 were retrieved for the present annotation review; this is not a claim that the collection is complete or that official and archival copies are unavailable. The mirror directory is

<https://www.gregsmithgolfcoach.com/wp-content/uploads/2014/04/>:

- `newsletter1.pdf` (Nov 2007) through `newsletter9.pdf` (Jan 2013), sequentially numbered.
- `TrackMan_Newsletter_2014.pdf` is issue #10 (Jan 2014); the filename differs from the numbered examples. This does not establish an end date for the series.
- **#7 (Oct 2010), “Ten Fundamentals”** presents historical vendor rules of thumb: driver launch weights of 85% toward the face in both planes, 75% for irons, and nominal spin decay of 4% per second. These are not universal material or aerodynamic constants. Its scalar loft-minus-attack definition requires a common-plane interpretation; spatial spin loft is the angle between the face normal and velocity.
- **#9 (Jan 2013), pp. 3 and 5** specifies a maximum-compression reporting reference with averaging qualifications. It lists TrackMan III/IIIe *absolute* specifications, described by the manufacturer as 95% confidence: speed ± 1.5 mph, attack $\pm 1.0^\circ$, path $\pm 1.0^\circ$, loft $\pm 0.8^\circ$, face $\pm 0.6^\circ$. Its separate repeatability column is not interchangeable with absolute error. Conditions include a centre of gravity within 10 mm of the geometric centre for attack/path, impact entirely on the face for loft/face, and successful measurement of the required club, ball-direction and spin data. A reporting reference does not mean every output is an instantaneous direct measurement; see also the dated club-data definitions above.
- **#8 (Jun 2011), p. 4** gives a 12.7 mm heel offset and a 2° face-angle change toward closed. Its universal-driver wording should not be adopted as a geometric law: the change depends on local face curvature. For an ideal circular horizontal section, $\Delta\theta = \arcsin(x/R)$ for lateral offset x and radius R , relative to the central normal. At $x = 12.7$ mm, radii of 250 and 500 mm give about 2.91° and 1.46° , respectively. These are geometric examples, not measured driver specifications; the sign depends on the heel/toe convention.

Caution

Those accuracy figures are manufacturer specifications. Leach et al. tested TrackMan Pro IIIe and GC2+HMT against an optical reference in an indoor, eight-golfer, 240-shot study (Leach et al. 2017, <https://doi.org/10.1016/j.measurement.2017.08.009>). In Table 6, 45% of TrackMan path differences and 66% of face-angle differences fall within $\pm 1^\circ$ of the *median difference*, not of zero error. Such a band describes dispersion after centering; a systematic bias can remain. It is not a direct pass/fail test of the vendor’s absolute limits.

Data availability is another quantity: TrackMan returned club velocity on 98% of shots, but

the other four club parameters on 62% overall and 19% for the utility wedge. The authors also describe installation compromises, possible radar interference, markers, and differences in reference point or impact timing. Reference calibration alone does not make all compared outputs the same measurand. Report bias, dispersion, uncertainty and return rate separately, and retain these device/setup limits when applying the result to a golfer's swing or to a newer monitor.

Foresight Sports

- GCQuad product/technology:
<https://www.foresightsports.com/products/gcquad-launch-monitor>
- Ball and club data (spherical correlation, fiducials):
<https://foresightsports.eu/ball-club-data/>
- HMT head measurement (dot-count data tiers):
<https://www.foresightsports.com/hmt-head-measurement>
- Club marker application guide:
<https://help.foresightsports.com/hc/en-us/articles/4408197030035>
- Marker troubleshooting:
<https://help.foresightsports.com/hc/en-us/articles/4405916455443>
- Design history (Popular Science): <https://www.popsoci.com/gear/foresight-sports-quadmax-photometric-golf-launch-monitor-development/>

Other Vendors

- FlightScope X3 announcement (The Golf Wire): <https://thegolfwire.com/326713-2/>
- Garmin R10 third-party explainers (My Golf Simulator):
<https://mygolfsimulator.com/garmin-r10-data/>,
<https://mygolfsimulator.com/garmin-r10-accuracy/>
- Garmin R50 (three-camera): <https://www.garmin.com/en-US/p/736810/>
- Full Swing KIT technology and specs:
<https://www.fullswinggolf.com/kit-launch-monitor-technology/>,
<https://www.fullswinggolf.com/kit-specs/>
- Rapsodo MLM2PRO spin (MyGolfSpy): <https://mygolfspy.com/news-opinion/rapsodo-mlm2pro-takes-spin-measurement-to-pro-level/>
- SkyTrak+ / ST+ vendor product and measurement description:
<https://www.skytrakgolf.com/products/skytrak-plus-launch-monitor>
- Uneekor EYE XO / Dimple Optix vendor description (Canadian site):
<https://uneekor.com/en-ca/products/eye-xo-standalone>
- Uneekor calibration guide (Carl's Place):
<https://www.carlofet.com/blog/calibrating-uneekor-launch-monitors>; official PDF:
https://download.uneekor.com/docs/EYEXO2_Calibration_Guide.pdf
- ProTee VX retailer review (PlayBetter):
<https://www.playbetter.com/blogs/golf-simulator-reviews/protee-vx-review>

A.4 Research Papers, Standards, and Technical Notes

Journal and proceedings papers, preprints, theses, author notes and test standards have different review and validation roles. Agreement with another monitor is not automatically accuracy against an independent reference; repeatability is not trueness. A cross-sport method needs golf-specific validation before its reported performance can be transferred.

- Penner, “The physics of golf,” Rep. Prog. Phys. 66 (2003):
<https://iopscience.iop.org/article/10.1088/0034-4885/66/2/202>
- Leach et al. 2017 launch-monitor validation (*Measurement*):
<https://www.sciencedirect.com/science/article/abs/pii/S0263224117305079>; open-access:
https://repository.lboro.ac.uk/articles/journal_contribution/9562799
- TrackMan 4 indoor reliability (J. Sports Sci. 2024):
<https://www.tandfonline.com/doi/full/10.1080/02640414.2024.2314864>
- Mevo+ vs TrackMan 4 agreement (2025):
<https://www.sciencedirect.com/science/article/pii/S2772696725000420>
- Bearman & Harvey 1976 golf ball aerodynamics:
<https://www.cambridge.org/core/journals/aeronautical-quarterly/article/abs/golf-ball-aerodynamics/67FE0903DB1CC12001F1ED1B1261C4B9>
- Smits & Smith aerodynamic model: <https://www.researchgate.net/publication/284037213>
- USGA Indoor Test Range conditions: <https://www.usga.org/content/dam/usga/images/equipment-standards/ITR-test-conditions-2028-ODS.pdf>
- Friction/tangential compliance and launch angle (MDPI 2020):
<https://doi.org/10.3390/proceedings2020049027>
- Wood, Henrikson and Broadie (2018), face/path influence study:
<https://doi.org/10.3390/proceedings2060249>. The player-test horizontal means (driver 0.76, 7-iron 0.69, wedge 0.61) use centre-strike and face-to-path filters; they are sample-specific, not universal coefficients or individual-shot error bounds.
- Cross oblique-impact golf experiments:
<https://www.physics.usyd.edu.au/~cross/GOLF/GOLF.htm>
- Cross & Nathan oblique collision formalism: <https://arxiv.org/pdf/1610.03464>
- Dynamic models review, Sports Engineering 25 (2022):
<https://link.springer.com/article/10.1007/s12283-022-00387-0>
- Stanford LES of golf ball flow (Sports Eng. 2019):
http://aero-comlab.stanford.edu/Papers/golf_ball_sports_engineering_2019.pdf
- Golf ball aerodynamics in still air (MDPI): <https://www.mdpi.com/2504-3900/2/6/238>
- Micro-Doppler of spinning projectiles on CW radar:
<https://www.researchgate.net/publication/317119893>
- SpinDOE camera spin estimation (transferable method): <https://arxiv.org/pdf/2303.03879>
- “Measuring Ball Spin by Image Registration”:
<https://www.researchgate.net/publication/2881136>

- Doppler ball-tracking thesis (WSU, TrackMan platform):
<https://baseball.physics.illinois.edu/trackman/JasonMartinThesisWSU.pdf>
- Alan Nathan's TrackMan (baseball) technical notes:
<https://baseball.physics.illinois.edu/trackman.html>
- Cochran & Stobbs, *Search for the Perfect Swing* (1968):
<https://archive.org/details/searchforperfect0000coch>

A.5 Engineering References (Tutelman et al.)

- 3D launch conditions from impact conditions:
<https://www.tutelman.com/golf/ballflight/3dlaunch.php>
- Smash factor: <https://www.tutelman.com/golf/ballflight/smashfactor.php>
- Gear effect: <https://www.tutelman.com/golf/ballflight/gearEffect1.php>
- Spin decay: <https://www.tutelman.com/golf/ballflight/spinDecay.php>
- Ball flight laws summary: <https://www.perfectgolfswingreview.net/ballflight.htm>
- TrackMan face-angle derivation discussion (Manzella forum): <https://forum.brianmanzellagolf.com/threads/how-does-trackman-flightscope-measure-the-club-face-angle.16591/>

A.6 DIY / Open-Source

- PiTrac GitHub (GPL-2.0): <https://github.com/PiTracLM/PiTrac>
- PiTrac documentation: <https://docs.pitrac.org/>
- PiTrac Hackaday.io project (design logs):
<https://hackaday.io/project/195042-pitrac-the-diy-golf-launch-monitor>
- OpenFlight upstream: <https://github.com/jewbetcha/openflight>
- OpenFlight Grafana write-up: <https://medium.com/grafana-labs/openflight-building-an-open-source-golf-launch-monitor-with-raspberry-pi-grafana-cloud-and-84d68ad40fc5>
- GC2-imitating OpenCV experiments: <https://github.com/ronheywood/opencv>
- GSA Golf simulator theory (marked-ball DIY methods):
<https://www.golf-simulators.com/GolfSimulatorTheory.html>

A.7 Hardware Datasheets, Protocols, and Standards

- OPS243 product brief:
https://omnipresense.com/wp-content/uploads/2024/01/OPS243-Product-Brief_004-F.pdf
- OPS243 AN-010 API interface:
https://omnipresense.com/wp-content/uploads/2025/10/AN-010-AD_API_Interface.pdf
- OPS243 AN-027 rolling buffer:
https://omnipresense.com/wp-content/uploads/2025/06/AN-027-A_Rolling-Buffer-1.pdf

- OPS243 AN-029 sports applications: https://omnipresense.com/wp-content/uploads/2026/06/AN-029-A_OPS243-for-Sports_260621.pdf
- RFbeam K-LD7 datasheet: <https://rfbeam.ch/product/k-ld7-radar-transceiver/>
- TI IWR6843 datasheet: <https://www.ti.com/lit/ds/symlink/iwr6843.pdf>; ISK EVM: <https://www.ti.com/tool/IWR6843ISK>; placement app note: <https://www.ti.com/document-viewer/lit/html/SWRA758>
- TI mmWave golf-swing forum thread: <https://e2e.ti.com/support/sensors-group/sensors/f/sensors-forum/1116424/iwr6843aop-golf-swing-analyzer-with-mmwave-radar>
- Raspberry Pi camera documentation (Global Shutter, XTR trigger): <https://www.raspberrypi.com/documentation/accessories/camera.html>
- GSPro Open Connect v1: <https://gsprogolf.com/GSProConnectV1.html>; community feedback: <https://github.com/tnbozman/gspro-interface/blob/main/OpenAPI-Documentation-Feedback.MD>; reference client: <https://github.com/travislang/gspro-garmin-connect-v2>
- USGA equipment standards: initial velocity TPX3007: <https://www.usga.org/content/dam/usga/pdf/Equipment/TPX3007-initial-velocity-test-procedure.pdf>; ODS TPX3006: <https://www.usga.org/content/dam/usga/pdf/Equipment/TPX3006-overall-distance-and-symmetry-test-procedure.pdf>; clubhead CT TPX3004 and MOI TPX3005 via <https://www.usga.org/equipment-standards.html>; 2028 ALC revision: <https://www.usga.org/content/usga/home-page/articles/2023/12/revised-golf-ball-testing-conditions-to-take-effect-in-2028.html>
- CFAR tutorial (CA/GO/SO/OS): <https://www.mathworks.com/help/phased/ug/constant-false-alarm-rate-cfar-detection.html>; Purdue lecture notes: https://engineering.purdue.edu/~mrb/resources/AltLectureF/Session_21.pdf
- Stalker baseball-spin patent (radar spin outside golf): <https://patents.google.com/patent/US10935657B2/en>
- Weibel Scientific radar object-detection patent: <https://patents.google.com/patent/EP1735637B1/en>

A.8 Comparative Testing and Market Analysis

These media, vendor and retailer accounts help locate comparisons and historical context. Before using a ranking or numerical accuracy claim, inspect its reference instrument, protocol, exclusions and uncertainty; the listing itself is not an endorsement of the test.

- MyGolfSpy indoor accuracy test: <https://mygolfspy.com/news-opinion/three-of-the-most-accurate-indoor-launch-monitors-and-one-to-avoid/>
- GCQuad vs TrackMan comparison article (Golf Simulator Zone): <https://golfsimulatorzone.com/gcquad-vs-trackman-accuracy-in-2026/>
- Foresight GCQuad/GC3 comparison: <https://www.foresightsports.com/blogs/golf-tips/comparing-golf-launch-monitor-foresight-sports-gcquad-and-gc3-stand-above-the-rest>
- Bushnell Launch Pro vs GC3 vs GCQuad: <https://www.playbetter.com/blogs/golf/bushnell-launch-pro-vs-gc3-vs-gc-quad-comparison>

- Photometric vs radar explainers: <https://www.playbetter.com/blogs/golf-simulator-comparisons/photometric-vs-radar-golf-launch-monitor>,
<https://golfbays.de/en/blogs/news/radar-vs-photometric-launch-monitors-indoors>
- Indoor flight compensation:
<https://www.cerogolf.com/post/how-launch-monitors-compensate-for-indoor-ball-flight>
- TrackMan history (Golf Monthly, Tuxen profile):
<https://www.golfmonthly.com/features/from-tracking-missiles-to-tracking-golf-balls-meet-the-man-that-you-didnt-realise-changed-golf>

Appendix B

Detailed Implementation Guidance

This appendix turns the review into engineering guidance at the level of signal-processing parameters, algorithms, and procedures, citing the governing sources throughout. Coordinate conventions follow Chapter 2.

B.1 Radar Signal Chain (OPS243-A)

B.1.1 Doppler Scaling and Resolution Budget

At $f_c = 24.125$ GHz, Equation (4.1) gives 71.7 Hz per mph. With a 30 kHz I/Q sample rate, the unambiguous span is ± 15 kHz $\approx \pm 209$ mph — adequate for ball speeds to ~ 200 mph. Velocity resolution is set by observation time: a 128-sample window (4.27 ms) gives $\Delta f = 234$ Hz ≈ 3.3 mph per raw bin; zero-padding to 4096 (the current pipeline) interpolates the peak but does not add information. For *spin* work the window, not the padding, must grow: resolving sidebands at $f_{\text{spin}} = S/60$ (e.g. 50 Hz at 3,000 rpm) requires windows of $\gtrsim 40$ ms, during which a 150 mph ball decelerates and the central line *chirps* — so the practical estimator de-chirps first (track the central line per US8845442’s spectral-trace step [73], resample the phase to remove it), then measures the residual modulation.

B.1.2 Spin-Rate Comb Estimation

The patent-documented chain (Section 4.4.2) maps to this concrete pipeline [73, 64]:

1. STFT with 50–75% overlap over the post-impact 50–250 ms; track the ball ridge (max-SNR bin per frame with continuity constraint).
2. De-chirp: mix each frame down by the tracked ridge frequency so the ball line sits at DC.
3. Estimate the sideband comb spacing on the de-chirped spectrum via cepstrum or harmonic product spectrum — both are robust to missing harmonics; the patent’s *qualification* step (verify equal spacing across ≥ 3 consecutive frames, and consistency of the harmonic-number assignment) is what separates real spin from clutter [73].
4. Report spin only when the qualified comb persists; otherwise fall back to a flagged estimate from club priors (Section B.3) — the Garmin R10’s documented behavior [36].

Expected detection physics: sideband amplitude scales with ball-surface asymmetry [73]; range balls and clean urethane balls read weakly (the reason for Titleist RCT metal-tagged balls [38] and FlightScope’s metallic-dot stickers [19]). Logging which ball type produced each detection will quantify this for a given deployment.

Disclosure references: US8845442 and US9868044 describe distinct radar spin-processing approaches [73, 27]. The latter’s dielectric-lens explanation should be distinguished from its claims’ signal-processing requirements (Section 7.3). These references identify technical material to study, not technique-wide expiry dates or implementation clearance (Section 7.7).

B.1.3 Club-Speed Extraction

The pre-impact spectrum contains a velocity *smear* from heel (slow) to toe (fast), plus shaft returns below. TrackMan resolves this by reconstructing the head silhouette and reporting the geometric center [69]; a single-radar approximation that tracks a stable reference is: gate the last 30–50 ms before the trigger timestamp, form the velocity histogram of CFAR-passing bins, discard the bottom decile (shaft/hosel) and top decile (toe glint), and report a fixed percentile (median of the remainder). Validate against the loft-dependent smash ceiling of Section 3.2 ([71]); reject or flag shots exceeding it, which independent testing shows is the signature of toe-lock errors [31].

B.2 Angle Measurement (K-LD7) and the EKF

B.2.1 Interferometric Angles

The K-LD7’s two receive patches implement Equation (4.2) with a $\lambda/2$ -class baseline; per-bin phase comparison of the two ADC channels after the range-Doppler FFT yields one angle per detection — the same phase-monopulse principle as US10850179 [74] and the R10’s three-receiver array [37]. Two practical constraints from the commercial art: (i) angle error grows as SNR falls, so weight each detection by measured SNR; (ii) mechanical alignment dominates the error budget — a 1° mount error is a 1° bias on every launch direction, amplified $\sim 1.15\times$ into face angle (Equation (4.6)).

B.2.2 Recommended Filter

Implement the Section 8.3 smoother concretely as: state $\mathbf{x} = (\mathbf{p}, \mathbf{v})$ (add spin states later); process model Equation (8.1) with Smits–Smith coefficients [54]; measurements: OPS243-A radial speed ($\dot{r}$, high rate, low noise), K-LD7 angle+range detections (lower rate, SNR-weighted covariance). Run a forward EKF then an RTS (Rauch–Tung–Striebel) backward smoother over the 50–150 ms burst, and evaluate the smoothed state at the sound-trigger timestamp minus the acoustic delay — this back-extrapolation is how commercial radars report launch conditions robustly despite early-flight clutter [73, 25]. The innovation-gating step of the EKF doubles as the outlier rejector for multipath and club returns. The same filtering architecture extends from point tracking (the ball) to rigid-body tracking (the club) via the screw-theoretic formulation of Chapter E, which is the recommended estimation layer for the IWR6843 integration.

B.2.3 Alignment Calibration Procedure

Adopt the commercial patterns: (i) Uneekor-style floor chart — a printed target-line chart at known positions establishes the target-line azimuth for both K-LD7s [4]; (ii) FlightScope-style known-trajectory check — roll or swing balls along a surveyed line and verify reported directions (US10338209 uses a Doppler simulator for the same purpose [25]); (iii) record the mount pose in the session log so data from different setups is never silently mixed.

B.3 Derived Club Data, With Priors

Implement the D-plane module once, use it both ways (Sections 3.1 and 4.5.2):

- **Forward** (simulation/validation): Tutelman’s calibrated closed form, Equations (3.3) and (3.4), plus spin from Equation (3.7) [72].

- **Inverse** (reporting): face angle from Equation (4.6) with w_f interpolated in dynamic loft between 0.87 (driver) and 0.75 (wedge) [61, 72]; dynamic loft analogously from launch angle and attack angle. Tag both as *derived* per Table 2.3.
- **Priors table** per club type (driver...wedge): static loft, typical dynamic-loft delta, spin-loft range, smash ceiling (Section 3.2), PGA/LPGA reference deliveries [67]. Used for: spin fallback estimates, outlier gating, and the smash sanity check.
- **Gear-effect bound**: when impact location is unknown, attach an uncertainty of up to $\pm 6^\circ$ of spin-axis tilt per 0.14in of possible driver miss [45, 70] to any derived face-to-path interpretation — and surface it in the UI rather than hiding it.

B.4 Optical Module (Phase 2) Design Parameters

The Wintriss disclosures provide optical measurement principles [29]; PiTrac provides an implementation reference [46]. The following choices are prototype parameters to verify against the selected hardware and capture conditions, rather than specifications fixed by a patent:

- **Sensor**: Raspberry Pi Global Shutter camera (IMX296, 1456×1088); global shutter is non-negotiable (rolling shutter skews a 150 mph ball by several pixels per row-time).
- **Strobed multi-exposure**: $N = 3\text{--}5$ IR pulses (850 nm, tens of μs each) inside one long exposure freeze N ball images per frame; at 150 mph the ball moves 6.7 cm/ms, so pulse spacing of $\sim 300\text{--}500 \mu\text{s}$ spaces images 2–3 cm apart in a 30 cm capture volume — matching the commercial capture geometry [11, 46].
- **Trigger**: reuse the SEN-14262 sound trigger; its $\sim 10 \mu\text{s}$ latency is far inside the strobe-timing budget, solving the triggering problem the Wintriss patent addressed with a microphone+radar pair [29].
- **Ball detection**: Hough circles on the strobed frame (PiTrac-proven [46]); depth from the known 42.67 mm diameter (the Wintriss mono-camera range cue [29]) or from a second camera via Equation (5.1).
- **Spin**: register the dimple texture between successive ball images over $SO(3)$ (Equation (5.2)); Gabor-filter pre-enhancement of dimples is documented in US12401909 [60]; with $\Delta t \approx 400 \mu\text{s}$, 3,000 rpm is only 7.2° of rotation — comfortably below the aliasing limit, and small enough that a local search around the D-plane-predicted axis converges quickly (Section 5.2.2). Surface-image registration appears in the Wintriss disclosures and Foresight’s technical account [29, 10]; neither source establishes blanket rights clearance for the prototype.
- **Impact location / face orientation** (later): either calibrated face-fiducial reconstruction, informed by the distinct Acushnet camera arrangements in US5501463 and US6758759 [17, 30], or an overhead second camera per the Uneekor geometry [75].

B.5 Flight Model and Validation Protocol

Ship Smits–Smith (Equation (8.2)) with spin decay [54], version the coefficients, and calibrate against measured trajectories using the US6186002 fitting approach [34] — outdoor sessions with the MLM2PRO (or simple carry ground-truth) provide the data. Validation against the MLM2PRO should follow the Leach protocol structure [31]: per-club shot blocks, Bland–Altman limits of

agreement per parameter, ball data compared directly, spin compared only on RPT/RCT balls (where the MLM2PRO's spin is camera-measured [38]), and club speed compared with an explicit reference-point caveat (Section 4.5). Log raw I/Q and (later) raw frames for every shot so algorithm changes can be replayed against history — the practice that made this review's accuracy analysis possible for the commercial units is exactly what an open project can do better.

Appendix C

Sensor Hardware and Integration Reference

This appendix is a specification-level reference for the commodity sensing components a launch monitor can be built from, drawn from vendor datasheets and application notes (all linked). The parts covered here are representative rather than prescriptive: they are the modules for which manufacturers publish enough detail to reason about performance, which makes them useful worked examples whether or not they end up in a given design.

Two of them are documented for this application by their own manufacturers. OmniPreSense’s sports application note AN-029 covers a golf configuration directly [40], which is unusual and worth exploiting — most radar modules are documented for traffic and presence sensing, leaving the sports-specific parameters to be derived from first principles.

C.1 OmniPreSense OPS243-A (24 GHz CW Doppler)

C.1.1 Hardware

Per the product brief [43]: 24.00–24.25 GHz ISM band, 11 dBm transmit power (FCC ID 2ALLL243A), patch antenna with **20° azimuth × 24° elevation** –3 dB beamwidth (footprint ~0.4 m wide at 1 m, 1.8 m at 5 m); motion detection 1–100 m; speed to 348 mph at 50 kps; accuracy spec 0.5%; USB CDC + 3.3 V UART (default 19,200 8N1); 5–24 V supply, 1.7 W active.

C.1.2 API Essentials

From AN-010 [41]: sample rate **S=n** (1–1000 kps; 10 kps default); buffer 1024/512/256/128 via **S>/S</S[/S(**; zero-padding **Xn/X=16/X=32** to a 4096-point FFT. Speed ceiling and resolution scale with sample rate (10 kps → 31.1 m/s at 0.061 m/s; 50 kps → 155.4 m/s at 0.304 m/s per 1024-sample buffer). Output modes: **OJ** JSON, **OT** timestamps, **OM** magnitudes, **O=n** multi-object (to 16), **OF** post-FFT, **OR** raw I/Q. Filters: **R>/R<** speed, **R±** direction, **M>** magnitude, **K+** peak averaging; built-in cosine-error correction **^/±n.n** (0–89°). **A!** persists settings to flash.

C.1.3 Rolling Buffer and Triggering

From AN-027 [42]: **G1** enters rolling-buffer mode with a fixed **4096-sample I/Q buffer organized as 32 segments of 128 samples**; trigger by software (**S!**) or a 3.3 V rising edge on **J3 pin 3 (HOST_INT)**; **S#n** sets the pre/post-trigger split (default 8 → 1024 pre + 3072 post). At 30 kps the buffer spans 136.5 ms with a 208.5 mph ceiling — AN-027’s “golf ball setting.” The app note wires a **SparkFun SEN-14262 Gate output directly to HOST_INT** (the standard hardware-trigger wiring) and flags the acoustic-latency budget: sound from 2 m arrives ~6.6 ms late, so the pre-trigger split must cover the impact-to-trigger gap (their example **S#18**).

C.1.4 The Vendor Golf Recipe (AN-029)

AN-029 [40] specifies: **S=30** (209 mph ceiling), **S<128** (128-sample segments), **X=32** (4096-point FFT, 0.1 mph resolution, ~ 200 Hz report rate), **US, R>10** to mask waggle, **M>10, 02** to report ball + club for smash factor (gated 1.0–1.50, matching Section 3.2), **K+** averaging, sensor 2–3 m behind the ball. Golf balls are rated “high” reflectivity, detectable 5–10 m.

C.2 RFbeam K-LD7 (24 GHz FSK, Dual-RX Angle)

Per the datasheet [52]: 24.050–24.250 GHz FSK (two frequencies, enabling range via phase difference); EIRP 6 dBm; 1 TX + **2 I/Q RX patches at 6.223 mm ($\approx \lambda/2$) spacing** — the interferometric baseline of Equation (4.2); beam $80^\circ \text{H} \times 34^\circ \text{V}$. Per-frame 256-point complex FFT; **angle $\pm 90^\circ$ at 1° resolution from the Rx1–Rx2 phase difference**; distance 5 cm–100 m (resolution 5 cm at the 5 m range setting); speed 0.1–100 km/h — note the **62 mph speed ceiling**, which confines the K-LD7 to angle and club-speed work; ball speed must come from a separate CW module. Frame time at the 100 km/h setting is 29 ms (~ 34 Hz).

UART protocol (115200 8E1 default, to 3 Mbaud via INIT): message types **RADC** (raw ADC: 256 I + 256 Q for Rx1@ f_A , Rx2@ f_A , Rx1@ f_B — the payload an interferometric angle solution consumes), **RFFT** (spectrum + threshold), **PDAT** (up to 12 raw targets: distance/speed/angle/magnitude), **TDAT** (tracked target), **DDAT** (flags), **DONE** (frame counter — use it to detect dropped frames); requested via the **GNFD** bitfield. Configuration: **RSPI** max speed, **RAI** max range, **THOF** threshold offset (10–60 dB), **RBFR** base frequency (three channels for multi-module coexistence — set co-located units to different channels), **TRFT** tracking filter, detection-window bounds (**MIRA/MARA/MIAN/MAAN/MISP/MASP**). Positive speed = receding.

C.3 TI IWR6843 (60–64 GHz FMCW MIMO)

Per the datasheet [57]: 60–64 GHz with **4 GHz chirp bandwidth** (~ 3.75 cm native range resolution); **3 TX / 4 RX = 12 virtual antennas** (TDM-MIMO); on-chip C674x DSP + radar hardware accelerator (FFT, log-magnitude, CFAR); complex-baseband ADC to 12.5 Msps. The LEVM/ISK-class EVMS give $\sim 120^\circ$ azimuth FoV with $\sim 15^\circ$ azimuth angular resolution (8 virtual azimuth antennas) and coarse elevation; the AOP variant trades resolution for a $130^\circ \times 130^\circ$ field. The out-of-box mmWave SDK demo streams a TLV point cloud (x, y, z , Doppler) at ~ 10 –20 Hz — *too slow for a 3 ms launch window*; using the IWR6843 for launch measurement requires a custom chirp/frame configuration (short frames, high Doppler span) and low-level processing, for which TI’s people-tracking labs (TIDEP-01000/01010) are the closest starting points [57]. TI’s forums confirm no golf-specific lab exists. The natural role: a single sensor that measures range, angle, and radial speed simultaneously — replacing a pair of single-baseline angle modules — once the custom chirp work is done.

C.4 Raspberry Pi Global Shutter Camera (Optical Module)

Sony IMX296 sensor: 1456×1088 , $3.45 \mu\text{m}$ pixels, true global shutter, exposures to $\sim 30 \mu\text{s}$, max ~ 60 fps streaming [51]. The key feature for Section B.4 is the **XTR external-trigger pad**: pulse low to expose (exposure = pulse width + $14.26 \mu\text{s}$); frame rate follows the pulse train; multiple cameras on one trigger line are hardware-synchronized; enable with `v4l2-ctl -c trigger_mode=1`

(early boards: remove R11 if Q2 is fitted) [51]. Continuous 60 fps cannot capture flight — the strobed multi-exposure design of Section B.4 is the correct use of this sensor.

C.5 Simulator Integration: GSPro Open Connect

GSPro’s Open Connect v1 [22] is the only fully documented open launch-monitor protocol and is the sensible native output (PiTrac ships it; E6/TruGolf and Foresight FSX require partnership agreements [46]). Mechanics: JSON over TCP to 127.0.0.1:0921, launch monitor as client. Minimum shot message: `DeviceID`, `ShotNumber`, `APIVersion`, `ShotDataOptions`{`ContainsBallData`, `ContainsClubData`}, and `BallData` with the five required fields:

Field	Units	Required	Maps to
<code>Speed</code>	mph	yes	ball speed (Table 2.2)
<code>VLA</code>	deg	yes	launch angle
<code>HLA</code>	deg	yes	launch direction
<code>TotalSpin</code>	rpm	yes*	spin rate
<code>SpinAxis</code>	deg	yes*	spin-axis tilt (– = draw)

*or `BackSpin+SideSpin`, related by Equation (2.2).

Optional `ClubData` carries `Speed`, `AngleOfAttack`, `Path`, `FaceToTarget`, `Loft`, `Lie`, `SpeedAtImpact`, `VerticalFaceImpact`, `HorizontalFaceImpact`, `ClosureRate` — precisely the Table 2.1 set, so the provenance tagging of Chapter 10 carries through unchanged. Status flags (`LaunchMonitorIsReady`, `LaunchMonitorBallDetected`, `IsHeartBeat`) and the 201 player-info response (handedness, selected club — useful for per-club priors, Section B.3) complete the loop.

C.6 Regulatory Constants for the Physics Engine

USGA/R&A equipment rules anchor the models of Chapters 3 and 8 [78]: ball mass ≤ 45.93 g and diameter ≥ 42.67 mm (the constants in Equation (8.1)); clubhead characteristic time $CT \leq 239$ μ s (+18 tolerance) $\approx$ COR 0.830 — the e in Equation (3.6); head MOI ≤ 5900 g cm² (+100) about the vertical CG axis — the upper bound on I_h in Equation (3.8); volume ≤ 460 cc. The Overall Distance Standard (317 yd + 3 at 120 mph clubhead / 10° / 2520 rpm; from January 2028, 125 mph / 11° / 2200 rpm) is a useful end-to-end sanity envelope for the flight model: a conforming ball simulated at ALC conditions must not materially exceed 320 yd [78].

C.7 Detection Theory: CFAR Selection

A typical CFAR setting (SNR > 15 over a 150-bin DC mask) is a cell-averaging scheme. The radar literature [33] distinguishes: CA-CFAR (mean of training cells around the cell under test, guard cells excluded; threshold $\alpha \hat{P}_n$ with α set by the desired false-alarm probability) — optimal in homogeneous noise; and **OS-CFAR** (order statistic: the k -th ranked training cell) — robust when two targets sit close together, at ~ 0.5 –1 dB detection loss. The club-then-ball geometry of a golf shot is precisely the two-closely-spaced-targets case, so OS-CFAR is the better default for the impact window.

Appendix D

Patent Portfolio Compendium

This appendix organizes selected launch-monitor patent publications by technical topic and named organization. It includes US grants, published US applications and explicitly identified foreign or PCT documents. The links provide a route to the disclosures; they do not establish an exhaustive portfolio search or a commercial product’s implementation. TrackMan’s marking notice lists 47 US numbers at the October 1, 2026 check [65]; Uneekor’s product-specific marking list expressly notes that other products and applications may exist outside its table [76].

The priority column records the indexed priority year of each row’s leading document. It is distinct from that application’s filing year and the document’s publication year, and it does not assign a common priority to every related number in the notes. A claim’s entitlement to priority and a patent’s term require separate analysis. Descriptions identify technical subject matter rather than the complete limitations of every claim. Neither a matching title nor a shared inventor proves common ownership, a continuation relationship or use in a named product. *Also* denotes another technical reference, not identical claim scope or verified family membership. Publication of an application is a document type, not an assertion that prosecution remains pending.

Two attribution corrections surfaced by this sweep are worth flagging prominently: (i) the widely cited “radar + image data 3D tracking” publications US10596416 / US11697046 / US12128275 identify **Topgolf Sweden AB (Toptracer)**, not TrackMan; and (ii) Full Swing’s launch-monitor publication US2020/0147470A1 corresponds to the grant US11311789B2. The Garmin and USGA entries below separate the disclosed mechanism from unsupported product and company-portfolio attributions.

D.1 TrackMan A/S (Incl. Interactive Sports Games A/S)

Radar Fundamentals and Target-Line Deviation (2004–2011)

Patent	Subject	Prio.	Notes
US8085188B2	Deviation of launched projectile vs. image-designated target direction	2004	
US8912945B2	Radar measurements transformed to a launch-and-target coordinate frame	2004	
US9857459B2	Target-direction deviation using co-rotated camera and radar	2004	
US10473778B2	Radar trajectory correlated with an image feature and displayed	2004	
US10690764B2	Radar trajectory mapped into launch-and-target coordinates	2004	

Patent	Subject	Prio.	Notes
US9958527B2	Direction-of-arrival sensor: extra RX antenna resolves monopulse phase ambiguity	2011	the sparse-array geometry of Section 4.2

Spin Rate and Spin Axis

Patent	Subject	Prio.	Notes
US8845442B2	Spin rate via harmonic sidebands; spin axis via trajectory/Magnus inversion	2005	Also EP1698380B1
US9645235B2	Unmarked-ball spin frequency from radar harmonic spacing	2005	
US10393870B2	Radar harmonic spin estimation without ball-marking information	2005	
US10962635B2	Spin frequency from radar modulation harmonic spacing	2005	
US11143754B2	Spin-axis inference from trajectory accelerations and spin frequency	2005	
US10850179B2	Projected spin axis from multi-receiver Doppler components	2018	
US11446546B2	Spin-axis projection from angular positions of radar frequency components	2018	
US11938375B2	Spin-axis component from receiver-pair phase differences	2018	
US11673029B2	Embedded radar markers with a great-circle proximity constraint	2019	
US12179068B2	Interior radar-reflective strips with great-circle coverage	2019	
US12042698B2	Toppling frequency of non-spherical rotating objects	2018	

Club Impact From Image Features

Patent	Subject	Prio.	Notes
US10953303B2	Impact timing and location from tracked club-image points	2017	
US11439886B2	Image-based impact geometry; single-imager embodiment with optional shaft marker	2017	
US11612801B2	Image-tracked line angle evaluated at impact time	2017	
US12263393B2	Fix-line angle at impact; 3D reconstruction uses known club geometry	2017	

Radar+Camera Fusion, Calibration, Tracer, Range, Short Game

Patent	Subject	Prio.	Notes
US10052542B2	Coordinating radar + image data; overlay	2004	
US10471328B2	Radar/image trajectory and target-line overlay	2004	
US10989791B2	Fused radar range/rate + imager angles track	2016	
US11828867B2	Radar range and image angles combined with sensor separation geometry	2016	
US11619708B2	Calibration by comparing radar and image tracks that share radar range	2020	
US12517218B2	Sensor calibration by weighted 3D track-residual minimization	2020	
US11748985B2	Master clock; composite multi-moment images	2019	
US12067775B2	Clock-referenced composite imaging	2019	
US12586211B2	Motion-triggered changes to imager operating parameters	2023	
US9855481B2	Radar trajectory projected into sports-video images	2009	Also US10315093B2, US10441863B2, US11135495B2, US11291902B2
US10379214B2	Launch-location association; one- and two-radar embodiments	2016	also US11086005B2, US11921190B2, US12618962B2
US11452911B2	Imager/secondary-sensor correspondence; range-radar embodiment	2019	also US11986698B2
US12036465B2	Player/shot association from motion-sensor time and location	2021	
US12186643B2	Camera line-of-sight and terrain model for bouncing/rolling ball position	2021	
US12702912B2	Robotic camera calibration and terrain-based ball localization	2021	USPTO Gazette
US10444339B2	Bounce/slide/roll transitions from velocity or acceleration changes	2016	also US11079483B2, US11619731B2, US11946997B2
US11285367B2	Strategy simulation from player capability	2018	also US12109473B2
US11951372B2	Model-based shot-improvement estimates; mishit-filtering disclosure	2020	also US12539454B2
US12616891B2	Automated ball/strike, biometric strike zone	2021	Baseball

D.2 Topgolf Sweden AB (Toptracer / Protracer)

The publications in this section identify Topgolf Sweden AB and include camera, radar and combined tracking methods.

Patent	Subject	Prio.	Notes
US8077917B2	Enhancing sports video with trajectory imagery	2005	
US10596416B2	3D tracking: radar + image data	2017	also US11697046B2, US12128275B2
US10898757B1	3D trace from image motion and polynomial/exponential radar-speed fits	2020	also US11504582B2, US11883716B2, US12330020B2
US11335013B2	Motion-based pre-processing, virtual time sync	2020	also US11557044B2, US12322122B2
US11644562B2	Trajectory extrapolation, origin determination	2020	Also US11771957B1, US12121771B2
US11513208B2	Radar spin from radial-velocity variations about center velocity	2020	Also US12105184B2
US12544624B2	Spin-candidate selection from observed trajectory and probable club	2023	Model-dependent inference
US11964188B2	Position-dependent trajectory error estimates and sensor/shot placement guidance	2021	
US11995846B2	Tracking with unverified detections	2020	also US12361570B2
US11815618B2	Doppler radar coexistence	2022	also US12253622B2
US12206977B2	Predictive camera control	2021	
US12298326B2	Wind velocity estimation	2022	
US12594460B2	Blob management for projectile tracking	2023	

D.3 FlightScope / EDH (Henri Johnson)

Patent	Subject	Prio.	Notes
WO2003032006A1	Path speed from directional velocity and angular deviation	2001	PCT publication
US8189857B2	Bounce-mark detection + tracking (cricket)	2007	
US9036864B2	Trajectory and bounce position	2011	
US9868044B2	Spin rate from periodic phase-demodulated radar features	2013	
US10775492B2	Spin axis from perpendicular receiver pairs	2013	
US10338209B2	Camera/radar calibration and tracking through missing data	2015	also US11016188B2

Patent	Subject	Prio.	Notes
US11573082B2	Tracking in varied environmental conditions	2019	
US12528005B2	Weather-based range prediction, club selector	2023	
US20160306036A1	Putting-green tracking	2013	Published application
US20180239012A1	Antenna with boresight optical system	2013	Published application

D.4 Full Swing Golf

Patent	Subject	Prio.	Notes
US11311789B2	CW/FMCW radar and non-uniform antenna array; grant of US2020/0147470A1	2018	Also US11844990B2
US11875517B2	Frame-difference tracking and calibrated golf-surface impact location	2020	also US12354282B2
US8758103B2	Fixed-plane crossing and image-based translation/rotation; IR embodiment	2009	also US9616346B2, US11033826B2
US8926416B2	Simulator: spin via image analysis	2007	also US10058733B2
US8414408B2	Ball-permeable screen, ball return	2009	also US8834284B2

Caution: US10605910B2/US11086008B2 (Alphawave Golf) and US11565166B2 (individual) surface in “Full Swing” text searches but are unrelated assignees.

Historical Light-Curtain Method: US5333874A (indexed priority 1992) describes two receiver planes and a post-screen rebound measurement used with stored spin values. Kiraly and Wintriss appear among the inventors; shared inventorship does not establish Foresight ownership. The index lists an individual original assignee and Full Swing as current assignee, rather than a Foresight portfolio entry. This is a rebound-based inference, not direct image tracking of a rotating ball surface.

D.5 Garmin

US11351436B2 — “Hybrid golf launch monitor” (indexed priority 2019): two laterally separated cameras whose capture timing depends on clubhead speed and a swing trigger. Claim 1 describes the two-camera arrangement; claim 9 adds a radar-based trigger and clubhead speed estimate. This disclosure does not establish that the Approach R10 implements the claimed arrangement. Other listed Garmin disclosures are US8647214B2 (2008, motion-sensor swing analysis) and US7467060B2 (2006, motion-parameter estimation). These entries do not establish the completeness of Garmin’s portfolio.

D.6 Rapsodo Pte. Ltd.

Patent	Subject	Prio.	Notes
US9955126B2	Core camera+radar moving-object analysis	2015	
US11170513B2	Marked-ball spin via surface-template matching	2016	
US11747461B2	Radar+camera data fusion	2018	
US20210299540A1	3D reconstruction of the launch scene	2018	Published application
US20230065614A1	Neural-network spin estimation from radar features	2021	Published application
US20230364468A1	Deep-learning ball/swing parameters from radar+image	2021	also club-side US20230070986A1
US12169941B1	Target-plane crossing localization	2024	
US12586248B2	Camera/radar moving-object tracking	2024	also US12548194B2
US12158517B1	Range-gated imager	2024	

D.7 Camera Vendors: Foresight/Wintriss, Creatz/Uneekor, Golfzon

Foresight Sports, Wintriss and WAWGD Disclosures

Patent	Subject	Prio.	Notes
US7292711B2	Image-based ball motion and surface-feature spin disclosure	2002	
US7324663B2	Image rotation/correlation search and alignment correction	2002	
US7497780B2	Integrated trigger, capture, calculation and display	2006	
US7540500B2	Foldable monitor housing	2006	
US7641565B2	Ball-placement detection / auto-arm	2006	
US8951138B2	Club path and face orientation around impact	2012	
US9737757B1	Alignment-stick target alignment	2016	
US10639537B2	Range tracking fused with launch monitor strike	2018	

Creatz Inc. (Uneekor)

Patent	Subject	Prio.	Notes
US9448067B2	Multi-camera (unsynchronized) trajectory via projection-plane intersection	2011	
US9605960B2	Single-camera plane-intersection trajectory	2011	
US9752875B2	Exposure/gain control from ambient brightness	2011	
US10247553B2	Start sensor: ball state at first movement	2011	
US10587797B2	Ball-image brightness compensation for spin marks	2016	
US10776929B2	Dynamic ROI from predicted ball motion	2016	
US11191998B2	Speed/angle-based spin estimates correct or replace mark-derived spin	2018	
US12008770B2	Dimple-position comparison with lighting correction for spin	2020	

Uneekor's marking page also lists the four licensed Wintriss/Foresight patents (US7497780, US7292711, US7641565, US7324663) [76, 3].

Golfzon Co., Ltd. (Sensing Core)

Patent	Subject	Prio.	Notes
US9242158B2	Two-stage sensing → simulation (latency hiding)	2011	
US9333409B2	Ball-candidate 2D trajectories used to derive 3D motion	2011	Also US9333412B2
US9162132B2	Image-based ball recognition and shot detection on a hitting area	2011	
US9514379B2	Difference images and estimated club trajectory inform ball-spin estimation	2011	
US10045008B2	Unsynchronized stereo cross-acquisition	2011	
US11364428B2	Spin fit by trajectory iteration vs. observed positions	2017	
US12002222B2	Database-derived loft/backspin and motion-vector spin-axis inference	2016	
US12599828B2	Marker-constellation spin between frames	2021	
US12605593B2	Rolling predictive ROI	2021	
US12478849B2	Single-camera putting-mat sensing	2020	
US20230347209A1	Stereo impact position on club face	2020	Published application

D.8 Acushnet (Titleist) — Selected Optical Disclosures

Patent	Subject	Prio.	Notes
US4136387A	Optical impact/launch monitoring	1977	
US5471383A	Shutterable cameras + retroreflective markers	1992	
US5501463A	Stereo + club/ball dots: face, path, contact location	1992	
US6241622B1	Portable camera/lighting system and trajectory computation	1998	Also US6533674B1, US6616543B1, US7086955B2
US6500073B1	Image-derived launch conditions and computed trajectory	1992	
US6758759B2	Separate club and ball camera arrangements	2001	
US7143639B2	Integrated club/ball cameras, illumination and trigger layout	2004	Also US7395696B2, US8500568B2, US8556267B2
US10668350B2	3D-image reconstruction; light-field camera in claim 1	2017	

D.9 Others and Foundational Radar Art

United States Golf Association: US6186002B1 (indexed priority 1998) is assigned in the publication record to the USGA, not Acushnet. It describes fitting lift and drag coefficients from launches with specified initial conditions and timed ballistic light-screen crossings, then relating those coefficients to Reynolds number and spin ratio. It is a useful experimental-modeling reference, not evidence that arbitrary launch-monitor outputs uniquely identify the aerodynamic coefficients.

AccuSport: US7209576B2 (indexed priority 2002; original assignee AccuSport International Inc.) describes golf-ball image processing, including marked-surface spin estimation and image calibration.

SkyTrak / SkyHawke: US12515116B2 (2023, swing-tag + launch-monitor fusion display) is one relevant disclosure. This catalog does not establish the complete SkyHawke portfolio or the patent marking of every SkyTrak model. No verified entries for **Zelocity**, **GolfTek**, **Ernest Sports** or **ProTee United** are supplied here; that search limitation establishes neither the absence of patents nor the technical ancestry of their products. **Voice Caddie (Ucomm):** US10338212B2 (2014, portable Doppler swing/ball analyzer). **Sports Sensors (Dilz):** US6079269A (1997) describes a short-range sports-object speed sensor, including a ball-catching application; it should not be labeled the Swing Speed Radar product patent. US6898971B2 (2000) addresses short-range speed measurement, including clubhead speed; US8007367B2 (2005) explicitly addresses swing duration and clubhead speed. These are different measurement tasks.

Weibel Scientific: EP1735637B1 (2004) describes simultaneous CW and FMCW tracking with three receivers, phase-based angles and maintained track records. Calling it only a CW-speed method omits a material part of the arrangement. **Applied Concepts:** US10935657B2 (2019) estimates projectile spin from radar spectral shoulder frequencies adjacent to a peak. Its relevance extends beyond golf; the disclosure is not evidence of a particular product's implemented accuracy.

Design implication

The catalog compares observations and assumptions behind similarly named outputs. Radar periodicity, image rotation and trajectory fitting need different calibration and ambiguity tests; a grant establishes no accuracy benchmark. For each implementation, retain the measured signals and model dependencies when reporting derived or estimated quantities. The rights boundaries in Section 7.7 apply throughout this appendix.

Appendix E

Radar Clubhead Kinematics: Motion and Observability

A useful club-motion model must connect what the sensor records to what a golfer wants to know: which point on the head is moving, how the face is turning, and how confidently those quantities are known near impact. Screw theory provides a compact representation of rigid motion, but choosing that representation does not make the motion observable. This appendix develops the measurement equations, exposes an exact ambiguity in instantaneous single-origin Doppler data, and specifies what an implementer must add and validate before reporting club-delivery parameters. The estimator described below is a research design, not a demonstrated IWR6843 golf instrument. The point-tracking framework in Section B.2 is a starting point; a rigid-body estimator also needs orientation, a calibrated head reference, and an explicit account of unresolved motion.

E.1 Context: Definitions Are Not an Estimator Specification

TrackMan defines club speed at the head’s geometric center immediately before first contact, while its path and attack-angle definitions use maximum compression as their time reference [69, 67]. These definitions specify intended outputs, not the complete sensing and estimation algorithm. The manufacturer’s discussion of a three-dimensional club silhouette and its patents describe additional information beyond one list of radial velocities; they do not establish that the particular Doppler-only solve below reconstructs a head pose [62, 74]. Nor does an algorithm’s appearance in a patent establish its implementation or independently measured accuracy in every product version. This appendix makes no claim about an undisclosed vendor state representation or the novelty or legal scope of the proposed design.

Instantaneous screw axes also have a distinct use in biomechanics. Vena et al. studied optically reconstructed pelvis, shoulder, and left-arm motion: Part 1 reports predominantly rotational contributions to marker velocity in those segments, and Part 2 reports the expected segment sequence in two of five subjects [79, 80]. Those observations neither validate a radar clubhead estimator nor establish a universal axis near the hands. The publisher abstracts and the visible Part 1 mathematical appendix were checked for this review; the subscription main texts were not accessed. The robotics conventions below are independently supported by the authors’ explanation of body and spatial twists [32]; Murray, Li, and Sastry remain a general reference [35].

E.2 Rigid Motion With Declared Frames and Reference Points

Assume a rigid clubhead over the interval being modeled, a stationary radar origin, and a right-handed Cartesian sensor frame. This is a head model: shaft bending and deformation during contact require separate states or an explicitly bounded approximation. Let O be a declared body-fixed reference point, $\mathbf{p}_O$ its sensor-frame position, $\mathbf{v}_O = \dot{\mathbf{p}}_O$ its velocity, and $\boldsymbol{\omega}$ the head’s angular velocity

expressed in the same frame. Use linear-first ordering throughout:

$$\xi_O = \begin{bmatrix} \mathbf{v}_O \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{R}^6. \quad (\text{E.1})$$

This six-component velocity description becomes a body or spatial Lie-algebra twist only after the reference and expression frame are specified below. For a body-fixed point whose instantaneous sensor-frame offset from O is $\mathbf{r}$,

$$\mathbf{v}(\mathbf{r}) = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}. \quad (\text{E.2})$$

Moving the reference to another body-fixed point Q therefore changes the linear component to $\mathbf{v}_Q = \mathbf{v}_O + \boldsymbol{\omega} \times (\mathbf{p}_Q - \mathbf{p}_O)$. The physical velocity field is unchanged; its six numerical coordinates are not invariant to changes of reference or frame.

For $\|\boldsymbol{\omega}\| > 0$, the instantaneous screw axis is the line

$$\begin{aligned} \mathbf{p}_{\text{axis}}(a) &= \mathbf{p}_O + \mathbf{r}_{\text{ISA}} + a\hat{\boldsymbol{\omega}}, & \mathbf{r}_{\text{ISA}} &= \frac{\boldsymbol{\omega} \times \mathbf{v}_O}{\|\boldsymbol{\omega}\|^2}, \\ h &= \frac{\boldsymbol{\omega} \cdot \mathbf{v}_O}{\|\boldsymbol{\omega}\|^2}, & \hat{\boldsymbol{\omega}} &= \frac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}. \end{aligned} \quad (\text{E.3})$$

Here a is distance along the axis and h is signed axial translation per radian. Substitution into Equation (E.2) gives velocity $h\boldsymbol{\omega}$ at every point on the axis, which verifies the construction. Pure translation has no unique finite axis, rest has no preferred axis, and near-zero angular speed makes the axis and pitch numerically sensitive. An instantaneous axis is not a uniquely defined swing plane: its moving line need not remain normal to one plane, and a head-point trajectory need not be planar. A plane estimate needs its own declared point trajectory, fitting interval, residual, and uncertainty. Low pitch or an axis near the grip can be hypotheses to test; neither is an assumption that the cited segment study licenses for every club swing.

E.2.1 Body and Spatial Twists for Pose Integration

Let $T = \begin{bmatrix} R & \mathbf{p}_O \\ 0 & 1 \end{bmatrix}$ map body coordinates to the sensor frame, with $R \in SO(3)$. For a linear-first pair $V = (\mathbf{v}, \boldsymbol{\omega})$, define $V^\wedge = \begin{bmatrix} [\boldsymbol{\omega}]_\times & \mathbf{v} \\ 0 & 0 \end{bmatrix}$, where $[\mathbf{a}]_\times \mathbf{b} = \mathbf{a} \times \mathbf{b}$. Then

$$\begin{aligned} V_b &= (R^\top \mathbf{v}_O, R^\top \boldsymbol{\omega}), & V_b^\wedge &= T^{-1} \dot{T}, \\ V_s &= (\mathbf{v}_O - \boldsymbol{\omega} \times \mathbf{p}_O, \boldsymbol{\omega}), & V_s^\wedge &= \dot{T} T^{-1}. \end{aligned} \quad (\text{E.4})$$

These are related by the adjoint of T [32]. A constant body-twist step uses $T_{k+1} = T_k \exp(V_{b,k}^\wedge \Delta t)$; a constant spatial-twist step uses $T_{k+1} = \exp(V_{s,k}^\wedge \Delta t) T_k$. For varying motion these are integration approximations; the increments must be composed in time order. Inserting sensor-frame $\mathbf{v}_O$ directly into the right-multiplying body update generally gives the wrong trajectory. Likewise, a logarithm of two measured poses describes a finite interframe displacement; dividing it by the interval is not automatically an unaliased instantaneous angular velocity.

E.3 The Measurement Model and Its Exact Ambiguity

First consider an ideal monostatic measurement represented by a single origin: a localized scattering point $\mathbf{p}_i \neq 0$ has sight line $\hat{\mathbf{u}}_i = \mathbf{p}_i / \|\mathbf{p}_i\|$, offset $\mathbf{r}_i = \mathbf{p}_i - \mathbf{p}_O$, and radial velocity z_i positive away

from the radar. Assume its Doppler return represents the instantaneous velocity of head material there. The conditional measurement equation is

$$z_i = \hat{\mathbf{u}}_i^\top \mathbf{v}_O + (\mathbf{r}_i \times \hat{\mathbf{u}}_i)^\top \boldsymbol{\omega} + \epsilon_i. \quad (\text{E.5})$$

With the ordering in Equation (E.1),

$$\mathbf{z} = A\boldsymbol{\xi}_O + \boldsymbol{\epsilon}, \quad A_i = [\hat{\mathbf{u}}_i^\top \mid (\mathbf{r}_i \times \hat{\mathbf{u}}_i)^\top]. \quad (\text{E.6})$$

Linearity holds conditional on the positions, associations, and rigid-motion model. It does not say that six independent unknowns can be recovered. Because $\mathbf{p}_i \times \hat{\mathbf{u}}_i = 0$, the same equation factors as

$$\begin{aligned} \mathbf{z} &= U(\mathbf{v}_O - \boldsymbol{\omega} \times \mathbf{p}_O) + \boldsymbol{\epsilon}, \\ A &= U[I \mid [\mathbf{p}_O]_\times], \quad \text{rank}(A) \leq 3, \end{aligned} \quad (\text{E.7})$$

where row i of U is $\hat{\mathbf{u}}_i^\top$. For any $\delta\boldsymbol{\omega}$, the perturbation $\delta\mathbf{v}_O = \delta\boldsymbol{\omega} \times \mathbf{p}_O$ leaves every predicted Doppler value unchanged. This is an exact three-dimensional null family even with arbitrarily many noncoplanar localized detections and noise-free measurements. Nearly parallel sight lines can further weaken the remaining three combinations; they do not explain away the structural nullspace. The observed combination is the translational component of V_s , not necessarily the speed of the head center.

Key point

One origin's instantaneous Doppler rows cannot determine the complete club velocity field under this model. Better signal-to-noise ratio or more rows at that same origin cannot remove this ambiguity. Range and angle histories, known feature geometry, independent orientation measurements, or sufficiently informative additional sensor geometry can supply information outside these rows; their contribution must appear in the actual measurement model.

For example, known and correctly associated noncollinear body features with resolved three-dimensional positions can constrain pose; their time history can then constrain motion. Unlabeled specular detections are not automatically such features. Two separated monostatic stations can generically raise the Doppler-only rank to five, but retain rotation about their baseline as a null direction with its corresponding translational change. Three noncollinear station origins can generically reach six with sufficient point/sight-line diversity. These are geometric possibilities, not guarantees of adequate precision. A multiple-input multiple-output (MIMO) array also requires its actual transmit/receive path model: a bistatic path derivative involves both ray directions. Twelve virtual array channels are not twelve independent monostatic stations, and the single-origin bound is not a theorem that every near-field, coherent, or feature-tracking radar architecture is limited to three observable motion coordinates.

E.3.1 What Least Squares and Covariance Actually Provide

For fixed A and zero-mean measurement error with known positive-definite covariance R_z , generalized least squares minimizes $(\mathbf{z} - A\boldsymbol{\xi})^\top R_z^{-1}(\mathbf{z} - A\boldsymbol{\xi})$. If A has full column rank, its estimator covariance is $(A^\top R_z^{-1} A)^{-1}$ under that model. This inverse does not exist for the six-unknown single-origin problem. A pseudoinverse selects a convention for the unobserved components; a zero eigenvalue in a pseudoinverse covariance is not evidence of zero uncertainty there. Report the identifiable combinations and the remaining nullspace, or explicitly state the extra observations or priors that define a full posterior.

Inspect singular values after noise whitening and declared parameter scaling, for example $R_z^{-1/2}AD$ with D converting dimensionless unknowns into chosen linear- and angular-velocity scales. Raw singular values mix meters per second with radians per second, so an unexplained threshold is not a physical observability criterion. Uncertain positions make A uncertain; position–Doppler correlations, calibration bias, ambiguous associations, and model discrepancy require joint propagation or an appropriate errors-in-variables model. Robust losses such as Huber or Cauchy generally require iterative fitting and their own uncertainty treatment. They do not turn an incorrect scattering model into a calibrated rigid-body measurement.

E.4 What Representative Hardware Adds

E.4.1 OPS243-A: Radial-Speed Information

The manufacturer’s OPS243 product brief distinguishes the Doppler-only A model from the range-capable C model [43]. Without per-return range and direction, the A output alone cannot assemble the located rows above. A Doppler spectrum is a scattering-amplitude-weighted signal, not a uniform sample of clubhead material velocities. Its peak need not always be the toe, its midpoint need not be the geometric-center speed, and its width is not a uniquely calibrated angular speed. For a stated example of 30 kps, 128 samples, and 24 GHz, $\Delta v = cf_s/(2f_cN) \approx 1.46$ m/s, or 3.27 mph, before accounting for the spectral window. Zero-padding samples the spectrum more finely but does not add the observation time needed to resolve closer velocities. A percentile or spectral-width feature may be useful after empirical calibration across aspect, club, speed, and clutter; it should retain that provenance.

E.4.2 K-LD7: Geometry and Operating Envelope

The K-LD7 Revision B datasheet specifies range, radial speed, and one angular coordinate, with a maximum speed setting of 100 km/h and a 29 ms cycle at that setting [52]. The speed limit is approximately 27.8 m/s, not a limit on total three-dimensional speed independent of viewing direction. A head moving at 50 m/s travels 1.45 m during that cycle; its radial component may alias even though another orientation does not. A single angular coordinate also does not localize arbitrary three-dimensional scatterers. Neither these specifications nor a frame’s angle output establishes that high-speed ball returns remain valid angle measurements after Doppler aliasing. Qualification would need the chosen geometry, waveform, association, and motion envelope.

E.4.3 IWR6843: A Coupled Waveform Budget

The IWR6843 has three transmitters, four receivers, a 60–64 GHz band, and a maximum specified ramp rate of 250 MHz/ μ s in Revision F [57]. Array angle estimation and multiplexing depend on antenna geometry and processing; they are not generically synonymous with monopulse [50]. Let T_c be the start-to-start chirp interval, n_T the number of cyclic time-division-multiplexed transmitters, and N the number of repeated samples *per transmitter*. For uniform slow-time sampling, wavelength λ , and a conventional unambiguous Doppler FFT,

$$T_r = n_T T_c, \quad |v_r| < \frac{\lambda}{4T_r}, \quad \Delta v = \frac{\lambda}{2NT_r}, \quad T_{\text{obs}} \approx NT_r. \quad (\text{E.8})$$

These follow from the round-trip phase increment $4\pi v_r T_r / \lambda$ and the sampled Fourier frequency grid [56, 50]. At the illustrative $\lambda = 5$ mm, $T_c = 25$ μ s, and $N = 128$:

Transmit Schedule	Speed Limit (m/s)	Δv (m/s)	T_{obs} (ms)
One transmitter	50.0	0.781	3.2
Three-transmitter TDM	16.7	0.260	9.6

The table is arithmetic, not a validated firmware configuration; frame idle time and processing can lower the output rate further. At the sampling boundary the signed velocity is ambiguous, so a 50 m/s limit provides no margin for a 50 m/s return. For three-transmitter TDM to exceed that unambiguous radial speed at this wavelength requires $T_c < 8.33 \mu\text{s}$. Sweeping 4 GHz already needs at least 16 μs at the stated maximum ramp rate, before settling and idle time. Thus those requirements cannot be combined in that simple waveform; ambiguity-resolution schemes or other multiplexing designs need their own analysis and validation.

Nominal range resolution is $c/(2B) \approx 3.75 \text{ cm}$ for an effective sampled bandwidth $B = 4 \text{ GHz}$ [56]. It does not guarantee three or four distinct returns from a head: projected radial extent, reflectivity, angular resolution, windowing, and detection thresholds also matter. Even over 3.2 ms, a 50 m/s radial return migrates 16 cm, more than four such bins. Acceleration, changing aspect, range migration, and inter-transmitter phase changes must therefore be included in the signal model or shown to be negligible for a bounded operating case. TI specifically describes motion-phase correction before TDM angle estimation [50]. Select sampled bandwidth, slope, ADC start/sample time, IF bandwidth, settling, transmitter schedule, output timing, and processing load together [56]; a bandwidth, frame rate, and velocity ceiling cannot be independently promised.

E.5 An Estimation Recipe With Explicit Acceptance Conditions

Step 1 — Preserve Geometry, Timing, and Detection Uncertainty

Retain the raw signal or sufficient diagnostics to check aliasing, motion compensation, and false detections. Choose CFAR training/guard cells and any order-statistic variant against recorded clutter and false-alarm requirements; no detector is universally preferable for every impact scene. For a declared right-handed sensor frame with range ρ , azimuth θ , and elevation ϕ , convert spherical coordinates to Cartesian positions:

$$\mathbf{p} = \rho \begin{bmatrix} \cos \phi \cos \theta & \cos \phi \sin \theta & \sin \phi \end{bmatrix}^T. \quad (\text{E.9})$$

Propagate the full range/angle covariance through this mapping and preserve relevant correlations with Doppler. A reported FFT bin spacing, a precision estimate, and demonstrated accuracy are different quantities. Assign timestamps to the acquisition intervals; a frame output timestamp is not an instantaneous sample of all returns.

Step 2 — Associate Returns With a Head Model

Use range, radial velocity, geometry, temporal continuity, and calibrated uncertainties jointly to distinguish head, shaft, ball, and body returns. A shaft point is not invariably slower in radial speed or nearer the sensor than the head, and speed alone cannot guarantee removal of body clutter. An SNR-weighted detection centroid can migrate as aspect changes; its derivative includes that migration and is not automatically the velocity of a body-fixed point. Choose O through a calibrated geometric model or identified body feature, and distinguish material-point velocity from apparent glint motion. If correspondence or head geometry is unresolved, retain multiple hypotheses or withhold outputs that depend on it.

Step 3 — Fit the Identifiable Quantities First

Use Equation (E.7) to fit the three-component combination $\mathbf{b} = \mathbf{v}_O - \boldsymbol{\omega} \times \mathbf{p}_O$ when the weighted sight-line matrix supports it. A smaller observable subspace may be all the data justify. Record residuals, scaled singular values, associations, and the acquisition interval. Do not publish an arbitrary six-component pseudoinverse solution as a fully measured twist. A full solve needs additional, explicitly modeled observations, such as calibrated feature-position histories or independent orientation information.

Step 4 — Establish Temporal Observability

A changing sight line is not a proof that smoothing recovers six arbitrary motion functions. For Doppler-only histories, even a constant rotation of every head position and velocity about the sensor preserves all radial histories while changing the target-relative motion; both trajectories can be smooth. Angle or feature-position observations may distinguish them, which is precisely why those observations must be included rather than used only as fixed coefficients in a velocity solve. Analyze the linearized multi-time measurement model with the chosen state transitions, geometry, calibration states, and priors. Report structural nullspaces separately from poor conditioning and from uncertainties controlled chiefly by a prior. Use withheld trajectories and deliberate near-degeneracies to test the claimed operating region.

Step 5 — Estimate Pose and Motion Consistently

An $SE(3)$ filter or batch smoother can combine pose, velocity, calibration, and feature observations using Equation (E.4). Feed raw residuals where practical, so the estimator retains their actual information and correlations; a per-frame diagnostic fit can still be useful. A backward smoother uses future measurements and must be labeled offline if they are unavailable at the desired reporting time. Select process uncertainty against observed dynamics, not an unsupported universal acceleration of one mile per hour per millisecond. Low-pitch or grip-proximity penalties are optional model assumptions whose sensitivity must be reported; they cannot resolve an unobservable quantity through measurement alone and become singular near zero angular speed if written using Equation (E.3) without safeguards. A prior that is weak in observed directions may completely determine an unobserved one.

Step 6 — Define the Impact Event and Model Boundary

Synchronize the radar, any optical reference, and the contact-event measurement, including sound-propagation and trigger latency where an acoustic trigger is used. Distinguish the last resolved pre-contact state, first contact, maximum compression, and separation. A prediction from the last pre-contact sample to another event is an extrapolated estimate with event-time and model uncertainty. The manufacturer's historical club-data explanation explicitly discusses using pre-impact data for an event-time definition [62]; the chosen product definition must accompany any comparison. Do not assume that every head is exactly rigid until a universal 500 μs switch, or extend a pre-contact process model through collision and turf contact without validation. Fit pre- and post-contact intervals separately if the purpose is to study collision-induced changes.

Step 7 — Report Only Supported Delivery Quantities

Let the target frame be right-handed with x down the target line, y to the golfer's left when looking down that line, and z upward. For a calibrated body-fixed point with body offset $\mathbf{r}_{gc}^b$,

first compute $\mathbf{v}_{gc} = \mathbf{v}_O + \boldsymbol{\omega} \times R\mathbf{r}_{gc}^b$ and express it in that target frame. Then speed is $\|\mathbf{v}_{gc}\|$, path positive to the right is $\text{atan2}(-v_y, v_x)$, and attack angle positive upward is $\text{atan2}(v_z, \sqrt{v_x^2 + v_y^2})$. Path is undefined for zero horizontal velocity, and the velocity direction is undefined at rest. A calibrated geometric-center offset, an anatomical grip point, and a changing radar centroid are not interchangeable.

Face closure needs a definition and orientation information. For a known target-frame unit face normal $\mathbf{n}$, rigid rotation gives $\dot{\mathbf{n}} = \boldsymbol{\omega} \times \mathbf{n}$. Its mathematical horizontal azimuth $\gamma = \text{atan2}(n_y, n_x)$ satisfies

$$\dot{\gamma} = \frac{n_x \dot{n}_y - n_y \dot{n}_x}{n_x^2 + n_y^2}, \quad n_x^2 + n_y^2 > 0. \quad (\text{E.10})$$

This generally differs from $\boldsymbol{\omega}$ projected onto the shaft axis. A right-positive face-angle convention reverses this azimuth sign; a label such as closing also needs the handedness and stated convention. A full head frame requires more orientation information than one normal, and face angle at the contact location may additionally require head geometry and impact location. Thus closure rate is not a free Doppler-only output or a parameter whose product availability can be declared universally exclusive.

Estimate swing plane and swing direction only through a specified trajectory-plane fit, with its residual and fit window. A low point requires an observed or modeled trajectory with $v_z = 0$ and a local height minimum; a vertical-velocity zero alone can be a maximum. A difference between pre- and post-impact angular velocities is a net kinematic change. Attributing it to gear effect requires an impact/contact model, timing, inertia, and consideration of continuing shaft/grip and other contact loads.

Propagate joint state, calibration, association, timing, and model uncertainty into each output and disclose unresolved directions or dependence on priors. Keep the book's distinction between sensor observations, quantities derived from them, and model-based estimates; merely changing the state representation promotes no output into a measured category. Kinematics alone also identifies neither golfer torques nor a control strategy: a hand-axis prior must not be presented as evidence that the golfer actually produced that motion by the assumed mechanism.

E.6 Validation That Can Falsify the Proposed Estimator

The following is a proposed qualification sequence, not a report of completed hardware or human testing. Use the release and evidence discipline of Chapter 11.

1. **Analytic and Synthetic Checks.** Verify the exact null family, frame conversions, pure-translation limit, and waveform arithmetic before fitting realistic data. Simulations should include noise, glint changes, aliasing, missed detections, correlated errors, and deliberately unobservable cases, and should test empirical interval coverage as well as point error. Recovering truth in a simulation generated with the same restrictive prior cannot establish that the prior holds for golfers.
2. **Independent Rig Reference.** Use synchronized calibrated encoders or optical tracking for pose and velocity on a rigid-motion rig. A fixed-axis pendulum supplies a known axis only if the bearings, mounting, head rigidity, and coordinate calibration support that model; its angular history is not known merely because it is called a pendulum. Test translations, multiple axes, low angular speeds, varying aspects, and moving-pivot cases as well as the easy planar case.

3. **Independent Cross-System Comparison.** Match the head point, event time, coordinate convention, club, environment, and uncertainty of the reference. A commercial monitor is a comparator, not automatically ground truth; unavailable or model-derived comparator outputs cannot validate an independent direct measurement. Report bias, limits of agreement, repeatability, missing-output rates, and uncertainty coverage against predefined requirements, including difficult cases.
4. **Mechanics Consistency With Stated Assumptions.** Check geometric identities only where their planar or rigid-body assumptions apply. An identity satisfied by construction is a software consistency check, not independent physical validation. Do not discard shots merely because a simplified smash-factor or swing-plane model is violated: inspect reference-point mismatch, contact, alignment, and model inadequacy, and preserve the rejection rule and excluded cases.

Design implication

The value of a screw representation is a common account of rigid motion and point transport. Its practical benefit depends on calibrated geometry, sufficient observations, consistent timing, and validated uncertainty. Declare what the data constrain before reporting a complete club state, and keep the inferred motion separate from claims about the forces or intentions that produced the swing.

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